

IE64 Processor User's Manual

IE64 64-bit RISC Processor Reference Manual

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1. Architecture Overview

The IE64 is a 64-bit RISC load-store CPU. It uses a clean, regular instruction encoding with the following core characteristics:

- **Word size:** 64-bit registers, 64-bit data path
- **Instruction width:** Fixed 8 bytes (64 bits) per instruction
- **Byte order:** Little-endian throughout (instruction encoding, memory access, immediates)
- **Architecture class:** Load-store-style register machine (integer computation is register-register/register-immediate; memory is accessed by explicit memory, stack, FPU-memory, and atomic instructions)

- **Integer condition model:** Compare-and-branch (no integer flags register)
- **Register file:** 32 general-purpose 64-bit registers (R0 hardwired to zero)
- **Address space:** 64-bit effective addresses for instruction fetch, explicit memory operations, stack operations, FPU memory operations, atomics, and MMU translation. IE64 exposes the active CPU-visible RAM size through the read-only CR_RAM_SIZE_BYTES control register.
- **Stack:** Full-descending, R31 serves as stack pointer. Hardware enforces 8-byte granularity for PUSH/POP. Software that requires wider call-boundary alignment must maintain it explicitly.
- **Interrupt model:** Legacy single-vector timer interrupt, plus MMU trap/vector registers when the MMU is enabled

2. Register File

The IE64 has 32 general-purpose 64-bit registers, addressed by a 5-bit field (0-31).

Register	Alias	Description
R0	--	Hardwired zero. Reads always return 0. Writes are silently discarded.
R1-R30	--	General-purpose registers. 64-bit read/write.
R31	SP	Stack pointer. Used implicitly by PUSH, POP, JSR, RTS, RTI, and interrupt entry. Initialised to 0x9F000 on reset.

Reset clears all general-purpose registers, then sets both R31 and R30 to STACK_START (0x9F000). R31 is the architectural stack pointer; no native IE64 instruction gives R30 special semantics.

Floating Point Registers (F0-F15): - 16 dedicated 32-bit registers for IEEE-754 single-precision floating point. - Accessed via dedicated FPU instructions (single-precision opcodes 0x60-0x7C and double-precision opcodes 0x80-0x90). - Initialised to 0.0 on reset.

Program Counter (PC): - 64-bit internal register, not directly addressable. - Full 64-bit PC. The CPU can address the full active visible RAM exposed by CR_RAM_SIZE_BYTES. - Initialised to 0x1000 (PROG_START) on reset. - Most sequential instructions advance PC by 8 bytes. - Control-transfer, trap, return, fault, interrupt, and stopped-state instructions use the PC behaviour defined by their individual instruction entries.

There is no integer flags register. All integer conditional branches use explicit register-register comparison within the branch instruction itself. FPU condition state is held separately in FPSR.

3. Instruction Encoding

Every IE64 instruction is exactly 8 bytes (64 bits), encoded in little-endian byte order.

3.1 Byte-Level Format

Byte:	0	1	2	3	4	5	6	7
	+-----+-----+-----+-----+-----+-----+-----+							
	Opcode	Rd Sz X	Rs unused	Rt unused	imm32 (LE)			
	+-----+-----+-----+-----+-----+-----+-----+							
Bits:	[7:0]	[7:3][2:1][0]	[7:3][2:0]	[7:3][2:0]	[31:0]			

3.2 Field Definitions

Field	Byte	Bits	Width	Description
opcode	0	[7:0]	8	Instruction opcode
Rd	1	[7:3]	5	Destination register index (0-31)
Size	1	[2:1]	2	Operand size code
X	1	[0]	1	Operand mode: 0 = register Rt, 1 = immediate imm32
Rs	2	[7:3]	5	First source register index (0-31)
unused	2	[2:0]	3	Reserved; software should encode 0; the CPU ignores these bits
Rt	3	[7:3]	5	Second source register index (0-31)
unused	3	[2:0]	3	Reserved; software should encode 0; the CPU ignores these bits
imm32	4-7	[31:0]	32	32-bit immediate value (little-endian)

3.3 Field Extraction Formulas

```
rd   = byte1 >> 3           // upper 5 bits of byte 1
size = (byte1 >> 1) & 0x03  // bits [2:1] of byte 1
xbit = byte1 & 1            // bit [0] of byte 1
rs   = byte2 >> 3           // upper 5 bits of byte 2
rt   = byte3 >> 3           // upper 5 bits of byte 3
imm32 = bytes[4] | (bytes[5] << 8) | (bytes[6] << 16) | (bytes[7] << 24)
```

3.4 Encoding Helper

```
instr[0] = opcode
instr[1] = (rd << 3) | (size << 1) | xbit
instr[2] = rs << 3
instr[3] = rt << 3
instr[4..7] = imm32 (little-endian)
```

3.5 Size Codes

Code	Suffix	Width	Mask
0	.B	8-bit	val & 0xFF
1	.W	16-bit	val & 0xFFFF
2	.L	32-bit	val & 0xFFFFFFFF
3	.Q	64-bit	val (no mask)

If no size suffix is specified in assembly, the default is .Q (64-bit).

3.6 Third Operand Resolution

When X=0, the third operand is the value of register Rt: `operand3 = regs[rt]`

When X=1, the third operand is the immediate, zero-extended to 64 bits: `operand3 = uint64(imm32)`

4. Complete Instruction Reference

4.0 Instruction Entry Schema

Each instruction entry uses the following field schema.

Field	Meaning
Operation	Canonical mnemonic and brief operation class.
Assembler Syntax	Assembly-language form for the instruction encoding.
Attributes	Size, privilege level, memory access class, and fixed-width constraints.
Description	Instruction-specific behaviour, including register, memory, control-register, or trap state changes.
Condition Codes	Integer condition-code effect. IE64 has no integer flags register; entries that write FPU condition state describe that effect explicitly.
Instruction Format	Opcode byte and fixed fields within the 8-byte instruction.
Instruction Fields	Meaning of <code>Rd</code> , <code>Rs</code> , <code>Rt</code> , <code>size</code> , <code>X</code> , and <code>imm32</code> for the entry.
Exceptions	Faults or traps architecturally raised by the instruction.
Notes	Architectural aliases, operand restrictions, or compatibility details.

Notation used in the `Operation` column is source-neutral pseudocode: `Rd`, `Rs`, and `Rt` are the register fields from bytes 1-3; `imm32` is the little-endian immediate in bytes 4-7; `maskToSize(x, s)` keeps the low bits selected by the `size` field; and `signExtend32to64(x)` extends a 32-bit signed quantity to the 64-bit register width. Rows describe the architectural effect. Instruction aliases are described only when they name the same encoded CPU behaviour.

4.1 Data Movement

1. MOVE - `move.s Rd, Rs`

Operation: `Rd = maskToSize(Rs, s).`

Assembler Syntax: `move.s Rd, Rs.`

Attributes: Memory: N; Size: B/W/L/Q.

Description: The selected-size value in `Rs` is copied to `Rd`; bits outside the selected size are cleared.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = `0x01`.

Instruction Fields: Byte 0 holds opcode `0x01`. Byte 1 bits 7-3 select `Rd`; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 0 for the register form. Byte 2 bits 7-3 select source register `Rs`. Byte 3 and bytes 4-7 are reserved by this form and ignored by the processor.

Exceptions: None.

Notes: None.

2. MOVE - `move.s Rd, #imm`

Operation: `Rd = maskToSize(imm32, s).`

Assembler Syntax: `move.s Rd, #imm.`

Attributes: Memory: N; Size: B/W/L/Q.

Description: The immediate field is masked to the selected size and loaded into Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x01.

Instruction Fields: Byte 0 holds opcode 0x01. Byte 1 bits 7-3 select Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 1 for the immediate form. Bytes 2-3 are reserved by this form and ignored by the processor. Bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: None.

3. MOVT - movt Rd, #imm

Operation: $Rd = (Rd \& 0x00000000FFFFFFFF) \mid (imm32 \ll 32).$

Assembler Syntax: `movt Rd, #imm.`

Attributes: Memory: N; Size: --.

Description: The immediate field is placed in the upper 32 bits of Rd; the low 32 bits of Rd are preserved.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x02.

Instruction Fields: Byte 0 holds opcode 0x02. Byte 1 bits 7-3 select destination register Rd; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Bytes 2-3 are reserved by this instruction and ignored by the processor. Bytes 4-7 hold unsigned imm32 in little-endian order; this field supplies the new high 32 bits of Rd.

Exceptions: None.

Notes: None.

4. MOVEQ - moveq Rd, #imm

Operation: $Rd = \text{signExtend32to64}(imm32).$

Assembler Syntax: `moveq Rd, #imm.`

Attributes: Memory: N; Size: --.

Description: The 32-bit immediate is sign-extended to 64 bits and loaded into Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x03.

Instruction Fields: Byte 0 holds opcode 0x03. Byte 1 bits 7-3 select destination register Rd; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Bytes 2-3 are reserved by this instruction and ignored by the processor. Bytes 4-7 hold imm32 in little-endian order; the processor sign-extends this field to 64 bits.

Exceptions: None.

Notes: None.

5. LEA - lea Rd, disp(Rs)

Operation: $Rd = Rs + \text{signExtend32to64}(\text{imm32})$.

Assembler Syntax: lea Rd, disp(Rs).

Attributes: Memory: N; Size: --.

Description: The sign-extended immediate is added to Rs, and the 64-bit result is loaded into Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x04.

Instruction Fields: Byte 0 holds opcode 0x04. Byte 1 bits 7-3 select destination register Rd; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select base register Rs; byte 2 bits 2-0 and byte 3 are reserved by this instruction and ignored by the processor. Bytes 4-7 hold signed imm32 displacement in little-endian order.

Exceptions: None.

Notes: None.

4.2 Load/Store

6. LOAD - load.s Rd, (Rs)

Operation: $Rd = \text{zeroExtend}(\text{mem}[Rs], s)$.

Assembler Syntax: load.s Rd, (Rs).

Attributes: Memory: Read; Size: B/W/L/Q.

Description: The effective address Rs is read using the selected size, zero-extended, and loaded into Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x10.

Instruction Fields: Byte 0 holds opcode 0x10. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); byte 1 bit 0 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select base register Rs; byte 2 bits 2-0 and byte 3 are reserved by this instruction and ignored by the processor. Bytes 4-7 hold signed imm32 displacement in little-endian order; encode zero for (Rs).

Exceptions: If MMU translation is enabled, reads can trap with cause 0 (FAULT_NOT_PRESENT), cause 1 (FAULT_READ_DENIED), or cause 10 (FAULT_SKAC). After optional MMU translation, physical backing is checked; a physical read outside implemented CPU-visible memory raises cause 0.

Notes: None.

7. LOAD - load.s Rd, disp(Rs)

Operation: $Rd = \text{zeroExtend}(\text{mem}[Rs + \text{signExtend}(\text{disp})], s)$.

Assembler Syntax: load.s Rd, disp(Rs).

Attributes: Memory: Read; Size: B/W/L/Q.

Description: The effective address $Rs + \text{signExtend}(\text{disp})$ is read using the selected size, zero-extended, and loaded into Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x10.

Instruction Fields: Byte 0 holds opcode 0x10. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); byte 1 bit 0 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select base register Rs; byte 2 bits 2-0 and byte 3 are reserved by this instruction and ignored by the processor. Bytes 4-7 hold signed imm32 displacement in little-endian order; encode zero for (Rs).

Exceptions: If MMU translation is enabled, reads can trap with cause 0 (FAULT_NOT_PRESENT), cause 1 (FAULT_READ_DENIED), or cause 10 (FAULT_SKAC). After optional MMU translation, physical backing is checked; a physical read outside implemented CPU-visible memory raises cause 0.

Notes: None.

8. STORE - store.s Rd, (Rs)

Operation: $\text{mem}[\text{Rs}] = \text{maskToSize}(\text{Rd}, s).$

Assembler Syntax: store.s Rd, (Rs).

Attributes: Memory: Write; Size: B/W/L/Q.

Description: The selected-size value in Rd is written to the effective address Rs.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x11.

Instruction Fields: Byte 0 holds opcode 0x11. Byte 1 bits 7-3 select source register Rd; bits 2-1 select size (B/W/L/Q); byte 1 bit 0 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select base register Rs; byte 2 bits 2-0 and byte 3 are reserved by this instruction and ignored by the processor. Bytes 4-7 hold signed imm32 displacement in little-endian order; encode zero for (Rs).

Exceptions: If MMU translation is enabled, writes can trap with cause 0 (FAULT_NOT_PRESENT), cause 2 (FAULT_WRITE_DENIED), or cause 10 (FAULT_SKAC). After optional MMU translation, physical backing is checked; a physical write outside implemented CPU-visible memory raises cause 0.

Notes: None.

9. STORE - store.s Rd, disp(Rs)

Operation: $\text{mem}[\text{Rs} + \text{signExtend}(\text{disp})] = \text{maskToSize}(\text{Rd}, s).$

Assembler Syntax: store.s Rd, disp(Rs).

Attributes: Memory: Write; Size: B/W/L/Q.

Description: The selected-size value in Rd is written to the effective address $\text{Rs} + \text{signExtend}(\text{disp})$.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x11.

Instruction Fields: Byte 0 holds opcode 0x11. Byte 1 bits 7-3 select source register Rd; bits 2-1 select size (B/W/L/Q); byte 1 bit 0 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select base register Rs; byte 2 bits 2-0 and byte 3 are reserved by this instruction and ignored by the processor. Bytes 4-7 hold signed imm32 displacement in little-endian order; encode zero for (Rs).

Exceptions: If MMU translation is enabled, writes can trap with cause 0 (FAULT_NOT_PRESENT), cause 2 (FAULT_WRITE_DENIED), or cause 10 (FAULT_SKAC). After optional MMU translation, physical backing is checked; a physical write outside implemented CPU-visible memory raises cause 0.

Notes: None.

Memory Access Widths

Size	Transfer Width	Architectural Memory Access
.B	1 byte	One 8-bit little-endian access
.W	2 bytes	One 16-bit little-endian access
.L	4 bytes	One 32-bit little-endian access
.Q	8 bytes	One 64-bit little-endian access

64-bit memory access: .Q loads and stores transfer one little-endian 64-bit quantity at the effective address. The CPU ISA defines the effective address, transfer width, and register result. Address validity, translation, protection, and fault behaviour are defined by the CPU memory and MMU rules.

4.3 Arithmetic

10. ADD - add.s Rd, Rs, Rt

Operation: $Rd = \text{maskToSize}(Rs + Rt, s)$.

Assembler Syntax: add.s Rd, Rs, Rt.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor adds Rt and Rs, masks the sum to the selected size, and writes it to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x20.

Instruction Fields: Byte 0 holds opcode 0x20. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 0 for the register form. Byte 2 bits 7-3 select source register Rs; byte 3 bits 7-3 select source register Rt. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this form and ignored by the processor.

Exceptions: None.

Notes: None.

11. ADD - add.s Rd, Rs, #imm

Operation: $Rd = \text{maskToSize}(Rs + \text{imm32}, s)$.

Assembler Syntax: add.s Rd, Rs, #imm.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor adds the immediate field and Rs, masks the sum to the selected size, and writes it to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x20.

Instruction Fields: Byte 0 holds opcode 0x20. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 1 for the immediate form. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0 and byte 3 are reserved by this form and ignored by the processor. Bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: None.

12. SUB - sub.s Rd, Rs, Rt

Operation: $Rd = \text{maskToSize}(Rs - Rt, s)$.

Assembler Syntax: sub.s Rd, Rs, Rt.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor subtracts Rt and Rs, masks the difference to the selected size, and writes it to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x21.

Instruction Fields: Byte 0 holds opcode 0x21. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 0 for the register form. Byte 2 bits 7-3 select source register Rs; byte 3 bits 7-3 select source register Rt. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this form and ignored by the processor.

Exceptions: None.

Notes: None.

13. SUB - sub.s Rd, Rs, #imm

Operation: $Rd = \text{maskToSize}(Rs - \text{imm32}, s)$.

Assembler Syntax: sub.s Rd, Rs, #imm.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor subtracts the immediate field and Rs, masks the difference to the selected size, and writes it to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x21.

Instruction Fields: Byte 0 holds opcode 0x21. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 1 for the immediate form. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0 and byte 3 are reserved by this form and ignored by the processor. Bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: None.

14. MULU - mulu.s Rd, Rs, Rt

Operation: $Rd = \text{maskToSize}(Rs * Rt, s)$ (unsigned).

Assembler Syntax: mulu.s Rd, Rs, Rt.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor multiplies Rs by Rt as unsigned integers, masks the product to the selected size, and writes the result to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x22.

Instruction Fields: Byte 0 holds opcode 0x22. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 0 for the register form. Byte 2 bits 7-3 select source register Rs; byte 3 bits 7-3 select source register Rt. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this form and ignored by the processor.

Exceptions: None.

Notes: None.

15. MULU - mulu.s Rd, Rs, #imm

Operation: $Rd = \text{maskToSize}(Rs * \text{imm32}, s)$ (unsigned).

Assembler Syntax: mulu.s Rd, Rs, #imm.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor multiplies Rs by the zero-extended immediate as unsigned integers, masks the product to the selected size, and writes the result to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x22.

Instruction Fields: Byte 0 holds opcode 0x22. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 1 for the immediate form. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0 and byte 3 are reserved by this form and ignored by the processor. Bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: None.

16. MULS - muls.s Rd, Rs, Rt

Operation: $Rd = \text{maskToSize}(\text{int64}(Rs) * \text{int64}(Rt), s)$ (signed).

Assembler Syntax: muls.s Rd, Rs, Rt.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor multiplies Rs by Rt as signed integers, masks the product to the selected size, and writes the result to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x23.

Instruction Fields: Byte 0 holds opcode 0x23. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 0 for the register form. Byte 2 bits 7-3 select source register Rs; byte 3 bits 7-3 select source register Rt. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this form and ignored by the processor.

Exceptions: None.

Notes: None.

17. MULS - muls.s Rd, Rs, #imm

Operation: $Rd = \text{maskToSize}(\text{int64}(Rs) * \text{int64}(\text{uint64}(\text{imm32})), s)$ (signed).

Assembler Syntax: muls.s Rd, Rs, #imm.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor multiplies Rs by the zero-extended imm32 converted to int64, applies the selected size only when masking the product written to Rd, and writes the result to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x23.

Instruction Fields: Byte 0 holds opcode 0x23. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 1 for the immediate form. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0 and byte 3 are reserved by this form and ignored by the processor. Bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: None.

18. DIVU - divu.s Rd, Rs, Rt

Operation: If Rt == 0, Rd = 0; otherwise Rd = maskToSize(Rs / Rt, s) (unsigned).

Assembler Syntax: divu.s Rd, Rs, Rt.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor divides Rs by Rt as unsigned integers, masks the quotient to the selected size, and writes the result to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x24.

Instruction Fields: Byte 0 holds opcode 0x24. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 0 for the register form. Byte 2 bits 7-3 select source register Rs; byte 3 bits 7-3 select source register Rt. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this form and ignored by the processor.

Exceptions: None.

Notes: If the divisor is zero and Rd is not R0, Rd receives zero.

19. DIVU - divu.s Rd, Rs, #imm

Operation: If imm32 == 0, Rd = 0; otherwise Rd = maskToSize(Rs / imm32, s) (unsigned).

Assembler Syntax: divu.s Rd, Rs, #imm.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor divides Rs by the zero-extended immediate as unsigned integers, masks the quotient to the selected size, and writes the result to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x24.

Instruction Fields: Byte 0 holds opcode 0x24. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 1 for the immediate form. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0 and byte 3 are reserved by this form and ignored by the processor. Bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: If the divisor is zero and Rd is not R0, Rd receives zero.

20. DIVS - divs.s Rd, Rs, Rt

Operation: If $Rt == 0$, $Rd = 0$; otherwise $Rd = \text{maskToSize}(\text{int64}(Rs) / \text{int64}(Rt), s)$ (signed).

Assembler Syntax: `divs.s Rd, Rs, Rt.`

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor divides Rs by Rt as signed integers, masks the quotient to the selected size, and writes the result to Rd .

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x25.

Instruction Fields: Byte 0 holds opcode 0x25. Byte 1 bits 7-3 select destination register Rd ; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 0 for the register form. Byte 2 bits 7-3 select source register Rs ; byte 3 bits 7-3 select source register Rt . Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this form and ignored by the processor.

Exceptions: None.

Notes: If the divisor is zero and Rd is not $R0$, Rd receives zero. DIVS interprets Rs and the register divisor as full 64-bit signed integers and applies the selected size only when masking the quotient written to Rd . For `.b`, `.w`, and `.l`, the operands are not sign-extended from the selected width before division.

21. DIVS - divs.s Rd, Rs, #imm

Operation: If $\text{imm32} == 0$, $Rd = 0$; otherwise $Rd = \text{maskToSize}(\text{int64}(Rs) / \text{int64}(\text{uint64}(\text{imm32})), s)$ (signed).

Assembler Syntax: `divs.s Rd, Rs, #imm.`

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor divides Rs by the zero-extended imm32 converted to int64 , applies the selected size only when masking the quotient written to Rd , and writes the result to Rd .

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x25.

Instruction Fields: Byte 0 holds opcode 0x25. Byte 1 bits 7-3 select destination register Rd ; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 1 for the immediate form. Byte 2 bits 7-3 select source register Rs ; byte 2 bits 2-0 and byte 3 are reserved by this form and ignored by the processor. Bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: If the divisor is zero and Rd is not $R0$, Rd receives zero. DIVS interprets Rs as a full 64-bit signed integer and the immediate divisor as zero-extended imm32 converted to int64 ; the selected size is applied only when masking the quotient written to Rd . For `.b`, `.w`, and `.l`, the operands are not sign-extended from the selected width before division. MODS differs: it sign-extends both operands to the selected width before taking the remainder.

22. MOD - mod.s Rd, Rs, Rt

Operation: If $Rt == 0$, $Rd = 0$; otherwise $Rd = \text{maskToSize}(Rs \% Rt, s)$ (unsigned).

Assembler Syntax: `mod.s Rd, Rs, Rt.`

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor divides Rs by Rt as unsigned integers and writes the selected-size remainder to Rd .

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x26.

Instruction Fields: Byte 0 holds opcode 0x26. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 0 for the register form. Byte 2 bits 7-3 select source register Rs; byte 3 bits 7-3 select source register Rt. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this form and ignored by the processor.

Exceptions: None.

Notes: If the divisor is zero and Rd is not R0, Rd receives zero.

23. MOD - mod.s Rd, Rs, #imm

Operation: If $\text{imm32} == 0$, $Rd = 0$; otherwise $Rd = \text{maskToSize}(Rs \% \text{imm32}, s)$ (unsigned).

Assembler Syntax: mod.s Rd, Rs, #imm.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor divides Rs by the zero-extended immediate as unsigned integers and writes the selected-size remainder to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x26.

Instruction Fields: Byte 0 holds opcode 0x26. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 1 for the immediate form. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0 and byte 3 are reserved by this form and ignored by the processor. Bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: If the divisor is zero and Rd is not R0, Rd receives zero.

24. NEG - neg.s Rd, Rs

Operation: $Rd = \text{maskToSize}(-\text{int64}(Rs), s)$.

Assembler Syntax: neg.s Rd, Rs.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor negates Rs as a signed integer, masks the result to the selected size, and writes it to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x27.

Instruction Fields: Byte 0 holds opcode 0x27. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); byte 1 bit 0 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: None.

Notes: None.

25. MODS - mods.s Rd, Rs, Rt/#imm

Operation: Let D be Rt for the register form or imm32 for the immediate form, sign-extended to the selected size. If $D == 0$, $Rd = 0$; otherwise $Rd = \text{maskToSize}(\text{signExtend}(Rs, s) \% D, s)$.

Assembler Syntax: mods.s Rd, Rs, Rt/#imm.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor computes signed remainder using truncation-toward-zero division semantics and writes the selected-size result to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x28.

Instruction Fields: Byte 0 holds opcode 0x28. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 selects the third operand source: 0 selects Rt, 1 selects imm32. Byte 2 bits 7-3 select source register Rs. When bit 0 is 0, byte 3 bits 7-3 select source register Rt and bytes 4-7 are ignored. When bit 0 is 1, byte 3 is ignored and bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: If the divisor is zero and Rd is not R0, Rd receives zero.

26. MULHU - mulhu Rd, Rs, Rt/#imm

Operation: Upper 64 bits of unsigned Rs * operand3.

Assembler Syntax: mulhu Rd, Rs, Rt/#imm.

Attributes: Memory: N; Size: Q.

Description: The processor multiplies Rs by the third operand as unsigned integers and writes the upper 64 bits of the product to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x29.

Instruction Fields: Byte 0 holds opcode 0x29. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select quadword size (Q); bit 0 selects the third operand source: 0 selects Rt, 1 selects imm32. Byte 2 bits 7-3 select source register Rs. When bit 0 is 0, byte 3 bits 7-3 select source register Rt and bytes 4-7 are ignored. When bit 0 is 1, byte 3 is ignored and bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: None.

27. MULHS - mulhs Rd, Rs, Rt/#imm

Operation: Upper 64 bits of signed int64(Rs) * int64(operand3).

Assembler Syntax: mulhs Rd, Rs, Rt/#imm.

Attributes: Memory: N; Size: Q.

Description: The processor multiplies Rs by the third operand and writes the upper 64 bits of the signed 128-bit product to Rd. Register operands use the full 64-bit value in Rt; immediate operands use zero-extended imm32 converted to int64.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x2A.

Instruction Fields: Byte 0 holds opcode 0x2A. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select quadword size (Q); bit 0 selects the third operand source: 0 selects Rt, 1 selects imm32. Byte 2 bits 7-3 select source register Rs. When bit 0 is 0, byte 3 bits 7-3 select source register Rt and bytes 4-7 are ignored. When bit 0 is 1, byte 3 is ignored and bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: None.

Immediate operands: ALU immediates are zero-extended to 64 bits before the operation. For signed immediate forms this means #0xFFFFFFFF is the positive value 4294967295 for MULS/DIVS .q; use MOVEQ or a register operand when a negative signed operand is required. MODS sign-extends the selected operand width after this zero-extension, so .b/ .w/ .l immediates are interpreted at the selected width.

Division by zero: If the divisor (Rt or imm32) is zero, the result is 0 (no exception raised). This applies to DIVU, DIVS, MOD, and MODS.

NEG is a 2-operand instruction: it reads Rs and writes the two's complement negation to Rd.

4.4 Logical

28. AND - and.s Rd, Rs, Rt

Operation: $Rd = \text{maskToSize}(Rs \ \& \ Rt, \ s).$

Assembler Syntax: and.s Rd, Rs, Rt.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor performs a bitwise AND of Rs and Rt, masks the result to the selected size, and writes it to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x30.

Instruction Fields: Byte 0 holds opcode 0x30. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 0 for the register form. Byte 2 bits 7-3 select source register Rs; byte 3 bits 7-3 select source register Rt. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this form and ignored by the processor.

Exceptions: None.

Notes: None.

29. AND - and.s Rd, Rs, #imm

Operation: $Rd = \text{maskToSize}(Rs \ \& \ \text{imm32}, \ s).$

Assembler Syntax: and.s Rd, Rs, #imm.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor performs a bitwise AND of Rs and the immediate field, masks the result to the selected size, and writes it to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x30.

Instruction Fields: Byte 0 holds opcode 0x30. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 1 for the immediate form. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0 and byte 3 are reserved by this form and ignored by the processor. Bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: None.

30. OR - or.s Rd, Rs, Rt

Operation: $Rd = \text{maskToSize}(Rs \mid Rt, s).$

Assembler Syntax: or.s Rd, Rs, Rt.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor performs a bitwise OR of Rs and Rt, masks the result to the selected size, and writes it to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x31.

Instruction Fields: Byte 0 holds opcode 0x31. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 0 for the register form. Byte 2 bits 7-3 select source register Rs; byte 3 bits 7-3 select source register Rt. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this form and ignored by the processor.

Exceptions: None.

Notes: None.

31. OR - or.s Rd, Rs, #imm

Operation: $Rd = \text{maskToSize}(Rs \mid \text{imm32}, s).$

Assembler Syntax: or.s Rd, Rs, #imm.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor performs a bitwise OR of Rs and the immediate field, masks the result to the selected size, and writes it to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x31.

Instruction Fields: Byte 0 holds opcode 0x31. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 1 for the immediate form. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0 and byte 3 are reserved by this form and ignored by the processor. Bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: None.

32. EOR - eor.s Rd, Rs, Rt

Operation: $Rd = \text{maskToSize}(Rs \wedge Rt, s).$

Assembler Syntax: eor.s Rd, Rs, Rt.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor performs a bitwise exclusive-OR of Rs and Rt, masks the result to the selected size, and writes it to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x32.

Instruction Fields: Byte 0 holds opcode 0x32. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 0 for the register form. Byte 2 bits 7-3 select source register Rs; byte 3 bits 7-3 select source register Rt. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this form and ignored by the processor.

Exceptions: None.

Notes: None.

33. EOR - eor.s Rd, Rs, #imm

Operation: $Rd = \text{maskToSize}(Rs \oplus \text{imm32}, s)$.

Assembler Syntax: eor.s Rd, Rs, #imm.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor performs a bitwise exclusive-OR of Rs and the immediate field, masks the result to the selected size, and writes it to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x32.

Instruction Fields: Byte 0 holds opcode 0x32. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 1 for the immediate form. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0 and byte 3 are reserved by this form and ignored by the processor. Bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: None.

34. NOT - not.s Rd, Rs

Operation: $Rd = \text{maskToSize}(\sim Rs, s)$.

Assembler Syntax: not.s Rd, Rs.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The processor inverts Rs, masks the result to the selected size, and writes it to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x33.

Instruction Fields: Byte 0 holds opcode 0x33. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); byte 1 bit 0 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: None.

Notes: None.

NOT is a 2-operand instruction. It performs bitwise complement of Rs, masked to the specified size, and stores the result in Rd.

4.5 Shifts

35. LSL - lsl.s Rd, Rs, Rt

Operation: $Rd = \text{maskToSize}(Rs \ll (Rt \& 63), s)$.

Assembler Syntax: lsl.s Rd, Rs, Rt.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The shift count is taken from Rt masked to six bits. The processor shifts Rs left, masks the result to the selected size, and writes it to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x34.

Instruction Fields: Byte 0 holds opcode 0x34. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 0 for the register form. Byte 2 bits 7-3 select source register Rs; byte 3 bits 7-3 select source register Rt. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this form and ignored by the processor.

Exceptions: None.

Notes: None.

36. LSL - lsl.s Rd, Rs, #imm

Operation: $Rd = \text{maskToSize}(Rs \ll (\text{imm32} \& 63), s).$

Assembler Syntax: lsl.s Rd, Rs, #imm.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The shift count is taken from the immediate field masked to six bits. The processor shifts Rs left, masks the result to the selected size, and writes it to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x34.

Instruction Fields: Byte 0 holds opcode 0x34. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 1 for the immediate form. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0 and byte 3 are reserved by this form and ignored by the processor. Bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: None.

37. LSR - lsr.s Rd, Rs, Rt

Operation: $Rd = \text{maskToSize}(Rs \gg (Rt \& 63), s).$

Assembler Syntax: lsr.s Rd, Rs, Rt.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The shift count is taken from Rt masked to six bits. The processor shifts Rs right logically, masks the result to the selected size, and writes it to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x35.

Instruction Fields: Byte 0 holds opcode 0x35. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 0 for the register form. Byte 2 bits 7-3 select source register Rs; byte 3 bits 7-3 select source register Rt. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this form and ignored by the processor.

Exceptions: None.

Notes: None.

38. LSR - lsr.s Rd, Rs, #imm

Operation: $Rd = \text{maskToSize}(Rs \gg (\text{imm32} \& 63), s).$

Assembler Syntax: `lsr.s Rd, Rs, #imm.`

Attributes: Memory: N; Size: B/W/L/Q.

Description: The shift count is taken from the immediate field masked to six bits. The processor shifts Rs right logically, masks the result to the selected size, and writes it to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x35.

Instruction Fields: Byte 0 holds opcode 0x35. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 1 for the immediate form. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0 and byte 3 are reserved by this form and ignored by the processor. Bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: None.

39. ASR - asr.s Rd, Rs, Rt

Operation: $Rd = \text{maskToSize}(\text{signedRs} \gg (Rt \& 63), s).$

Assembler Syntax: `asr.s Rd, Rs, Rt.`

Attributes: Memory: N; Size: B/W/L/Q.

Description: The shift count is taken from Rt masked to six bits. The processor shifts Rs right arithmetically, masks the result to the selected size, and writes it to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x36.

Instruction Fields: Byte 0 holds opcode 0x36. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 0 for the register form. Byte 2 bits 7-3 select source register Rs; byte 3 bits 7-3 select source register Rt. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this form and ignored by the processor.

Exceptions: None.

Notes: None.

40. ASR - asr.s Rd, Rs, #imm

Operation: $Rd = \text{maskToSize}(\text{signedRs} \gg (\text{imm32} \& 63), s).$

Assembler Syntax: `asr.s Rd, Rs, #imm.`

Attributes: Memory: N; Size: B/W/L/Q.

Description: The shift count is taken from the immediate field masked to six bits. The processor shifts Rs right arithmetically, masks the result to the selected size, and writes it to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x36.

Instruction Fields: Byte 0 holds opcode 0x36. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 is encoded as 1 for the immediate form. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0 and byte

3 are reserved by this form and ignored by the processor. Bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: None.

41. CLZ - clz.l Rd, Rs

Operation: `Rd = LeadingZeros32(uint32(Rs)).`

Assembler Syntax: `clz.l Rd, Rs.`

Attributes: Memory: N; Size: L.

Description: The processor counts leading zero bits in the low 32 bits of Rs and writes the count to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x37.

Instruction Fields: Byte 0 holds opcode 0x37. Byte 1 bits 7-3 select destination register Rd. Byte 1 bits 2-1 are reserved by this instruction and ignored by the processor; the operation always uses the low 32 bits of Rs. Byte 1 bit 0 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: None.

Notes: None.

42. SEXT - sext.s Rd, Rs

Operation: Sign-extend byte/word/long source to 64 bits.

Assembler Syntax: `sext.s Rd, Rs.`

Attributes: Memory: N; Size: B/W/L/Q.

Description: The selected byte, word, or long source value is sign-extended to 64 bits and written to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x38.

Instruction Fields: Byte 0 holds opcode 0x38. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); byte 1 bit 0 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: None.

Notes: None.

43. ROL - rol.s Rd, Rs, Rt/#imm

Operation: Rotate left within the selected width.

Assembler Syntax: `rol.s Rd, Rs, Rt/#imm.`

Attributes: Memory: N; Size: B/W/L/Q.

Description: The selected-width value is rotated left and written to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x39.

Instruction Fields: Byte 0 holds opcode 0x39. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 selects the third operand source: 0 selects Rt, 1 selects imm32. Byte 2 bits 7-3 select source register Rs. When bit 0 is 0, byte 3 bits 7-3 select source register Rt and bytes 4-7 are ignored. When bit 0 is 1, byte 3 is ignored and bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: None.

44. ROR - ror.s Rd, Rs, Rt/#imm

Operation: Rotate right within the selected width.

Assembler Syntax: ror.s Rd, Rs, Rt/#imm.

Attributes: Memory: N; Size: B/W/L/Q.

Description: The selected-width value is rotated right and written to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x3A.

Instruction Fields: Byte 0 holds opcode 0x3A. Byte 1 bits 7-3 select destination register Rd; bits 2-1 select size (B/W/L/Q); bit 0 selects the third operand source: 0 selects Rt, 1 selects imm32. Byte 2 bits 7-3 select source register Rs. When bit 0 is 0, byte 3 bits 7-3 select source register Rt and bytes 4-7 are ignored. When bit 0 is 1, byte 3 is ignored and bytes 4-7 hold unsigned imm32 in little-endian order.

Exceptions: None.

Notes: None.

45. CTZ - ctz.l Rd, Rs

Operation: Rd = TrailingZeros32(uint32(Rs)).

Assembler Syntax: ctz.l Rd, Rs.

Attributes: Memory: N; Size: L.

Description: The processor counts trailing zero bits in the low 32 bits of Rs and writes the count to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x3B.

Instruction Fields: Byte 0 holds opcode 0x3B. Byte 1 bits 7-3 select destination register Rd. Byte 1 bits 2-1 are reserved by this instruction and ignored by the processor; the operation always uses the low 32 bits of Rs. Byte 1 bit 0 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: None.

Notes: None.

46. POPCNT - popcnt.l Rd, Rs

Operation: Rd = OnesCount32(uint32(Rs)).

Assembler Syntax: popcnt.l Rd, Rs.

Attributes: Memory: N; Size: L.

Description: The processor counts one bits in the low 32 bits of Rs and writes the count to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x3C.

Instruction Fields: Byte 0 holds opcode 0x3C. Byte 1 bits 7-3 select destination register Rd. Byte 1 bits 2-1 are reserved by this instruction and ignored by the processor; the operation always uses the low 32 bits of Rs. Byte 1 bit 0 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: None.

Notes: None.

47. BSWAP - bswap.l Rd, Rs

Operation: $Rd = \text{ReverseBytes32}(\text{uint32}(Rs))$.

Assembler Syntax: bswap.l Rd, Rs.

Attributes: Memory: N; Size: L.

Description: The processor reverses byte order in the low 32 bits of Rs and writes the result to Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x3D.

Instruction Fields: Byte 0 holds opcode 0x3D. Byte 1 bits 7-3 select destination register Rd. Byte 1 bits 2-1 are reserved by this instruction and ignored by the processor; the operation always uses the low 32 bits of Rs. Byte 1 bit 0 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: None.

Notes: None.

Shift amount masking: The shift count is always masked to 6 bits (& 63), limiting the effective shift range to 0-63.

CLZ (Count Leading Zeros): A 2-operand instruction that counts the number of leading zero bits in the low 32 bits of Rs and stores the result in Rd. The result is an integer in the range 0-32: zero if the most-significant bit is set, 32 if the input is zero. Only the .l size suffix is supported. Writing to R0 is silently discarded (as with all instructions).

This instruction is useful for $O(1)$ floating-point normalisation, integer $\log_2$ computation, and highest-set-bit detection.

ASR sign extension: Before performing the arithmetic right shift, the source value is sign-extended according to the current size:

Size	Sign-extension
.B	$\text{int64}(\text{int8}(Rs))$
.W	$\text{int64}(\text{int16}(Rs))$
.L	$\text{int64}(\text{int32}(Rs))$
.Q	$\text{int64}(Rs)$

The result is then masked to the specified size after the shift.

LSL and LSR operate on the unsigned 64-bit value in Rs. The result is masked to the specified size.

ROL and ROR first mask the source to the selected size, rotate within that width, then mask the result again. The effective rotate count is masked to 3 bits for .B, 4 bits for .W, 5 bits for .L, and 6 bits for .Q.

CTZ, POPCNT, and BSWAP are 2-operand .l-only instructions. CTZ and POPCNT operate on uint32(Rs). BSWAP stores bits.ReverseBytes32(uint32(Rs)) zero-extended to 64 bits.

4.6 Floating Point (FPU)

The IE64 FPU provides native single-precision (f*) and double-precision (d*) IEEE-754 operations. Single-precision values use the 16 scalar registers f0-f15. Double-precision instruction encodings name the even-numbered F register of the even-odd register pair (f0, f2, ..., f14). The conceptual pair names below describe storage:

- d0 = f0:f1
- d1 = f2:f3
- d2 = f4:f5
- d3 = f6:f7
- d4 = f8:f9
- d5 = f10:f11
- d6 = f12:f13
- d7 = f14:f15

All d* mnemonics require even-numbered FP operands. Odd FP operand encodings are invalid. Writing a double clobbers both halves of the pair.

4.6.1 FPU Data Movement

48. FMOV - fmov fd, fs

Operation: fd = fs (FP copy).

Assembler Syntax: fmov fd, fs.

Attributes: Memory: none. Operand size: single precision. Privilege: unprivileged. FP operands: fd and fs encode f0-f15. FPSR/FPCR: neither FPSR nor FPCR is read or written.

Description: The processor copies the single-precision value in fs to fd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x60.

Instruction Fields: Byte 0 holds opcode 0x60. Byte 1 bits 7-3 select destination FP register fd; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register fs; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid fd or fs encoding enters the stopped processor state with PC unchanged.

Notes: None.

49. FLOAD - fload fd, disp(rs)

Operation: fd = mem32[rs + disp].

Assembler Syntax: fload fd, disp(rs).

Attributes: Memory: 32-bit read. Operand size: single precision. Privilege: unprivileged. FP operands: *fd* encodes *f0-f15*. Integer operand: *rs* supplies the base address. FPSR/FPCR: writes FPSR condition-code bits from the loaded value; FPCR is not read.

Description: The processor evaluates `mem32[rs + disp]` and writes the result to *fd*.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x61.

Instruction Fields: Byte 0 holds opcode 0x61. Byte 1 bits 7-3 select destination FP register *fd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select integer base register *rs*; byte 2 bits 2-0 and byte 3 are reserved by this instruction and ignored by the processor. Bytes 4-7 hold signed *imm32* displacement in little-endian order for the 32-bit memory transfer.

Exceptions: An invalid *fd* encoding enters the stopped processor state with PC unchanged. Read faults are those defined by the CPU memory and MMU rules.

Notes: None.

50. FSTORE - *fstore fs, disp(rs)*

Operation: `mem32[rs + disp] = fs`.

Assembler Syntax: `fstore fs, disp(rs)`.

Attributes: Memory: 32-bit write. Operand size: single precision. Privilege: unprivileged. FP operands: *fs* encodes *f0-f15*. Integer operand: *rs* supplies the base address. FPSR/FPCR: neither FPSR nor FPCR is read or written.

Description: The processor evaluates *fs* and writes the result to `mem32[rs + disp]`.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x62.

Instruction Fields: Byte 0 holds opcode 0x62. Byte 1 bits 7-3 select source FP register *fs*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select integer base register *rs*; byte 2 bits 2-0 and byte 3 are reserved by this instruction and ignored by the processor. Bytes 4-7 hold signed *imm32* displacement in little-endian order for the 32-bit memory transfer.

Exceptions: An invalid *fs* encoding enters the stopped processor state with PC unchanged. Write faults are those defined by the CPU memory and MMU rules.

Notes: None.

51. FMOVECR - *fmovecr fd, #idx*

Operation: `fd = ROM_Constant[idx]`.

Assembler Syntax: `fmovecr fd, #idx`.

Attributes: Memory: none. Operand size: single precision result from an 8-bit constant selector. Privilege: unprivileged. FP operands: *fd* encodes *f0-f15*. FPSR/FPCR: writes FPSR condition-code bits from the constant; FPCR is not read.

Description: The processor evaluates `ROM_Constant[idx]` and writes the result to *fd*.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x78.

Instruction Fields: Byte 0 holds opcode 0x78. Byte 1 bits 7-3 select destination FP register *fd*; byte 1 bits 2-0 and bytes 2-3 are reserved by this instruction and ignored by the processor. Bytes 4-7 hold unsigned *imm32* in little-endian order; the low 8 bits select the FPU constant index.

Exceptions: An invalid *fd* encoding enters the stopped processor state with PC unchanged.

Notes: None.

FLOAD/FSTORE always transfer 4 bytes (32 bits) between memory and an FP register. The *disp* is a signed 32-bit immediate. **FSTORE** uses *fs* as the source floating-point register and *rs* as the base integer register for the effective address.

FMOVECR loads a constant from the FPU ROM (indices 0-15). Indices outside this range load 0.0 and set the Z condition code.

Index	Constant	Index	Constant
0	Pi	8	1.0
1	e	9	2.0
2	log2(e)	10	10.0
3	log10(e)	11	100.0
4	ln(2)	12	1000.0
5	ln(10)	13	0.5
6	log10(2)	14	Smallest positive FP32 subnormal (0x00000001, approximately 1.40129846e-45)
7	0.0	15	FLT_MAX

4.6.2 FPU Arithmetic

52. FADD - fadd *fd*, *fs*, *ft*

Operation: $fd = fs + ft$.

Assembler Syntax: fadd *fd*, *fs*, *ft*.

Attributes: Memory: none. Operand size: single precision. Privilege: unprivileged. FP operands: *fd*, *fs*, and *ft* encode f0-f15. FPSR/FPCR: writes FPSR condition-code bits from the result and may set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates $fs + ft$ and writes the result to *fd*.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x63.

Instruction Fields: Byte 0 holds opcode 0x63. Byte 1 bits 7-3 select destination FP register *fd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register *fs*; byte 3 bits 7-3 select source FP register *ft*. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid *fd*, *fs*, or *ft* encoding enters the stopped processor state with PC unchanged.

Notes: None.

53. FSUB - fsub *fd*, *fs*, *ft*

Operation: $fd = fs - ft$.

Assembler Syntax: `fsub fd, fs, ft.`

Attributes: Memory: none. Operand size: single precision. Privilege: unprivileged. FP operands: `fd`, `fs`, and `ft` encode `f0-f15`. FPSR/FPCR: writes FPSR condition-code bits from the result and may set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates $fs - ft$ and writes the result to `fd`.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = `0x64`.

Instruction Fields: Byte 0 holds opcode `0x64`. Byte 1 bits 7-3 select destination FP register `fd`; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register `fs`; byte 3 bits 7-3 select source FP register `ft`. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid `fd`, `fs`, or `ft` encoding enters the stopped processor state with PC unchanged.

Notes: None.

54. FMUL - `fmul fd, fs, ft`

Operation: $fd = fs * ft$.

Assembler Syntax: `fmul fd, fs, ft.`

Attributes: Memory: none. Operand size: single precision. Privilege: unprivileged. FP operands: `fd`, `fs`, and `ft` encode `f0-f15`. FPSR/FPCR: writes FPSR condition-code bits from the result and may set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates $fs * ft$ and writes the result to `fd`.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = `0x65`.

Instruction Fields: Byte 0 holds opcode `0x65`. Byte 1 bits 7-3 select destination FP register `fd`; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register `fs`; byte 3 bits 7-3 select source FP register `ft`. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid `fd`, `fs`, or `ft` encoding enters the stopped processor state with PC unchanged.

Notes: None.

55. FDIV - `fdiv fd, fs, ft`

Operation: $fd = fs / ft$.

Assembler Syntax: `fdiv fd, fs, ft.`

Attributes: Memory: none. Operand size: single precision. Privilege: unprivileged. FP operands: `fd`, `fs`, and `ft` encode `f0-f15`. FPSR/FPCR: writes FPSR condition-code bits from the result and may set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates fs / ft and writes the result to `fd`.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = `0x66`.

Instruction Fields: Byte 0 holds opcode 0x66. Byte 1 bits 7-3 select destination FP register *fd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register *fs*; byte 3 bits 7-3 select source FP register *ft*. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid *fd*, *fs*, or *ft* encoding enters the stopped processor state with PC unchanged.

Notes: None.

56. FMOD - *fmod fd, fs, ft*

Operation: $fd = fs \% ft$.

Assembler Syntax: *fmod fd, fs, ft*.

Attributes: Memory: none. Operand size: single precision. Privilege: unprivileged. FP operands: *fd*, *fs*, and *ft* encode f0-f15. FPSR/FPCR: writes FPSR condition-code bits from the result and may set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates $fs \% ft$ and writes the result to *fd*.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x67.

Instruction Fields: Byte 0 holds opcode 0x67. Byte 1 bits 7-3 select destination FP register *fd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register *fs*; byte 3 bits 7-3 select source FP register *ft*. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid *fd*, *fs*, or *ft* encoding enters the stopped processor state with PC unchanged.

Notes: None.

57. FABS - *fabs fd, fs*

Operation: $fd = \lfloor fs \rfloor$.

Assembler Syntax: *fabs fd, fs*.

Attributes: Memory: none. Operand size: single precision. Privilege: unprivileged. FP operands: *fd* and *fs* encode f0-f15. FPSR/FPCR: writes FPSR condition-code bits from the result and does not set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates $\lfloor fs \rfloor$ and writes the result to *fd*.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x68.

Instruction Fields: Byte 0 holds opcode 0x68. Byte 1 bits 7-3 select destination FP register *fd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register *fs*; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid *fd* or *fs* encoding enters the stopped processor state with PC unchanged.

Notes: None.

58. FNEG - *fneg fd, fs*

Operation: $fd = -fs$.

Assembler Syntax: *fneg fd, fs*.

Attributes: Memory: none. Operand size: single precision. Privilege: unprivileged. FP operands: `fd` and `fs` encode `f0-f15`. FPSR/FPCR: writes FPSR condition-code bits from the result and does not set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates `-fs` and writes the result to `fd`.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = `0x69`.

Instruction Fields: Byte 0 holds opcode `0x69`. Byte 1 bits 7-3 select destination FP register `fd`; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register `fs`; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid `fd` or `fs` encoding enters the stopped processor state with PC unchanged.

Notes: None.

59. FSQRT - `fsqrt fd, fs`

Operation: `fd = sqrt(fs)`.

Assembler Syntax: `fsqrt fd, fs`.

Attributes: Memory: none. Operand size: single precision. Privilege: unprivileged. FP operands: `fd` and `fs` encode `f0-f15`. FPSR/FPCR: writes FPSR condition-code bits from the result and may set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates `sqrt(fs)` and writes the result to `fd`.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = `0x6A`.

Instruction Fields: Byte 0 holds opcode `0x6A`. Byte 1 bits 7-3 select destination FP register `fd`; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register `fs`; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid `fd` or `fs` encoding enters the stopped processor state with PC unchanged.

Notes: None.

60. FINT - `fint fd, fs`

Operation: `fd = round(fs)` (uses FPCR mode).

Assembler Syntax: `fint fd, fs`.

Attributes: Memory: none. Operand size: single precision. Privilege: unprivileged. FP operands: `fd` and `fs` encode `f0-f15`. FPSR/FPCR: reads FPCR rounding bits, writes FPSR condition-code bits from the rounded result, and does not set FPSR sticky exception flags.

Description: The processor rounds `fs` according to the rounding mode selected by FPCR and writes the single-precision result to `fd`.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = `0x6B`.

Instruction Fields: Byte 0 holds opcode `0x6B`. Byte 1 bits 7-3 select destination FP register `fd`; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register `fs`; byte 2 bits 2-0, byte 3, and

bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid *fd* or *fs* encoding enters the stopped processor state with PC unchanged.

Notes: None.

4.6.3 FPU Transcendentals

61. FSIN - *fsin fd, fs*

Operation: $fd = \sin(fs)$.

Assembler Syntax: *fsin fd, fs*.

Attributes: Memory: none. Operand size: single precision. Privilege: unprivileged. FP operands: *fd* and *fs* encode *f0-f15*. FPSR/FPCR: writes FPSR condition-code bits from the result and does not set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates $\sin(fs)$ and writes the result to *fd*.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x71.

Instruction Fields: Byte 0 holds opcode 0x71. Byte 1 bits 7-3 select destination FP register *fd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register *fs*; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid *fd* or *fs* encoding enters the stopped processor state with PC unchanged.

Notes: None.

62. FCOS - *fcos fd, fs*

Operation: $fd = \cos(fs)$.

Assembler Syntax: *fcos fd, fs*.

Attributes: Memory: none. Operand size: single precision. Privilege: unprivileged. FP operands: *fd* and *fs* encode *f0-f15*. FPSR/FPCR: writes FPSR condition-code bits from the result and does not set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates $\cos(fs)$ and writes the result to *fd*.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x72.

Instruction Fields: Byte 0 holds opcode 0x72. Byte 1 bits 7-3 select destination FP register *fd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register *fs*; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid *fd* or *fs* encoding enters the stopped processor state with PC unchanged.

Notes: None.

63. FTAN - *ftan fd, fs*

Operation: $fd = \tan(fs)$.

Assembler Syntax: *ftan fd, fs*.

Attributes: Memory: none. Operand size: single precision. Privilege: unprivileged. FP operands: *fd* and *fs* encode *f0-f15*. FPSR/FPCR: writes FPSR condition-code bits from the result and does not set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates $\tan(fs)$ and writes the result to *fd*.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x73.

Instruction Fields: Byte 0 holds opcode 0x73. Byte 1 bits 7-3 select destination FP register *fd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register *fs*; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid *fd* or *fs* encoding enters the stopped processor state with PC unchanged.

Notes: None.

64. FATAN - *fatan fd, fs*

Operation: $fd = \text{atan}(fs)$.

Assembler Syntax: *fatan fd, fs*.

Attributes: Memory: none. Operand size: single precision. Privilege: unprivileged. FP operands: *fd* and *fs* encode *f0-f15*. FPSR/FPCR: writes FPSR condition-code bits from the result and does not set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates $\text{atan}(fs)$ and writes the result to *fd*.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x74.

Instruction Fields: Byte 0 holds opcode 0x74. Byte 1 bits 7-3 select destination FP register *fd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register *fs*; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid *fd* or *fs* encoding enters the stopped processor state with PC unchanged.

Notes: None.

65. FLOG - *flog fd, fs*

Operation: $fd = \ln(fs)$.

Assembler Syntax: *flog fd, fs*.

Attributes: Memory: none. Operand size: single precision. Privilege: unprivileged. FP operands: *fd* and *fs* encode *f0-f15*. FPSR/FPCR: writes FPSR condition-code bits from the result and may set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates $\ln(fs)$ and writes the result to *fd*.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x75.

Instruction Fields: Byte 0 holds opcode 0x75. Byte 1 bits 7-3 select destination FP register *fd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register *fs*; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid *fd* or *fs* encoding enters the stopped processor state with PC unchanged.

Notes: None.

66. FEXP - fexp *fd*, *fs*

Operation: $fd = e^{fs}$.

Assembler Syntax: fexp *fd*, *fs*.

Attributes: Memory: none. Operand size: single precision. Privilege: unprivileged. FP operands: *fd* and *fs* encode *f0-f15*. FPSR/FPCR: writes FPSR condition-code bits from the result and may set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates e^{fs} and writes the result to *fd*.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x76.

Instruction Fields: Byte 0 holds opcode 0x76. Byte 1 bits 7-3 select destination FP register *fd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register *fs*; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid *fd* or *fs* encoding enters the stopped processor state with PC unchanged.

Notes: None.

67. FPOW - fpow *fd*, *fs*, *ft*

Operation: $fd = fs^{ft}$.

Assembler Syntax: fpow *fd*, *fs*, *ft*.

Attributes: Memory: none. Operand size: single precision. Privilege: unprivileged. FP operands: *fd*, *fs*, and *ft* encode *f0-f15*. FPSR/FPCR: writes FPSR condition-code bits from the result and may set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates fs^{ft} and writes the result to *fd*.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x77.

Instruction Fields: Byte 0 holds opcode 0x77. Byte 1 bits 7-3 select destination FP register *fd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register *fs*; byte 3 bits 7-3 select source FP register *ft*. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid *fd*, *fs*, or *ft* encoding enters the stopped processor state with PC unchanged.

Notes: None.

4.6.4 FPU Comparison and Conversion

68. FCMP - fcmp *rd*, *fs*, *ft*

Operation: $rd = (fs < ft ? -1 : (fs > ft ? 1 : 0))$.

Assembler Syntax: fcmp *rd*, *fs*, *ft*.

Attributes: Memory: none. Operand size: single precision. Privilege: unprivileged. FP operands: *fs* and *ft* encode *f0-f15*; *rd* receives the integer comparison result. FPSR/FPCR: writes FPSR condition-code bits and may set the IO sticky flag for unordered operands; FPCR is not read.

Description: The processor compares the single-precision values in *fs* and *ft*, writes -1, 0, or 1 to *rd*, and updates FPSR condition state. If either operand is NaN, the comparison is unordered: *rd* receives 0, the NaN condition code is set, and the IO sticky exception flag is set.

Condition Codes: See the FPU condition-state description for this instruction class.

Instruction Format: Fixed 8-byte instruction; opcode = 0x6C.

Instruction Fields: Byte 0 holds opcode 0x6C. Byte 1 bits 7-3 select destination integer register *rd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register *fs*; byte 3 bits 7-3 select source FP register *ft*. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid *fs* or *ft* encoding enters the stopped processor state with PC unchanged.

Notes: +Inf compares equal to +Inf and greater than finite values.

69. FCVTIF - *fcvtif fd, rs*

Operation: *fd* = float32(int32(*rs*)).

Assembler Syntax: *fcvtif fd, rs*.

Attributes: Memory: none. Operand size: single precision result from a 32-bit signed integer source. Privilege: unprivileged. FP operands: *fd* encodes *f0-f15*; *rs* is an integer source register. FPSR/FPCR: writes FPSR condition-code bits from the FP result; FPCR is not read.

Description: The processor evaluates float32(int32(*rs*)) and writes the result to *fd*.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x6D.

Instruction Fields: Byte 0 holds opcode 0x6D. Byte 1 bits 7-3 select destination FP register *fd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source integer register *rs*; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid *fd* encoding enters the stopped processor state with PC unchanged.

Notes: None.

70. FCVTFI - *fcvtfi rd, fs*

Operation: *rd* = int32(*fs*) (saturating).

Assembler Syntax: *fcvtfi rd, fs*.

Attributes: Memory: none. Operand size: single precision source to 32-bit signed integer result. Privilege: unprivileged. FP operands: *fs* encodes *f0-f15*; *rd* receives the integer result. FPSR/FPCR: may set the FPSR IO sticky flag for NaN or saturation; FPCR is not read.

Description: The processor converts *fs* to a signed 32-bit integer with saturation and writes the result to *rd*.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x6E.

Instruction Fields: Byte 0 holds opcode 0x6E. Byte 1 bits 7-3 select destination integer register *rd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register *fs*; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid *fs* encoding enters the stopped processor state with PC unchanged.

Notes: None.

71. FMOVI - *fmovi fd, rs*

Operation: `fd = bits_to_float(uint32(rs)).`

Assembler Syntax: *fmovi fd, rs.*

Attributes: Memory: none. Operand size: 32-bit bit pattern moved from integer to single precision FP storage. Privilege: unprivileged. FP operands: *fd* encodes *f0-f15*; *rs* is an integer source register. FPSR/FPCR: writes FPSR condition-code bits from the resulting FP bit pattern; FPCR is not read.

Description: The processor evaluates `bits_to_float(uint32(rs))` and writes the result to *fd*.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x6F.

Instruction Fields: Byte 0 holds opcode 0x6F. Byte 1 bits 7-3 select destination FP register *fd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source integer register *rs*; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid *fd* encoding enters the stopped processor state with PC unchanged.

Notes: None.

72. FMOV0 - *fmov0 rd, fs*

Operation: `rd = uint64(float_to_bits(fs)).`

Assembler Syntax: *fmov0 rd, fs.*

Attributes: Memory: none. Operand size: 32-bit FP bit pattern zero-extended to an integer result. Privilege: unprivileged. FP operands: *fs* encodes *f0-f15*; *rd* receives the integer result. FPSR/FPCR: neither FPSR nor FPCR is read or written.

Description: The processor evaluates `uint64(float_to_bits(fs))` and writes the result to *rd*.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x70.

Instruction Fields: Byte 0 holds opcode 0x70. Byte 1 bits 7-3 select destination integer register *rd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register *fs*; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid *fs* encoding enters the stopped processor state with PC unchanged.

Notes: None.

FCVTFI saturates to `INT32_MAX` or `INT32_MIN` on overflow. NaNs return 0 and set the IO exception flag.

4.6.5 FPU Status and Control

73. FMOVSR - fmovsr rd

Operation: rd = FPSR.

Assembler Syntax: fmovsr rd.

Attributes: Memory: none. Operand size: 32-bit status word. Privilege: unprivileged. Integer operand: rd receives FPSR, with writes to R0 discarded. FPSR/FPCR: reads FPSR; FPCR is not read or written.

Description: The processor evaluates FPSR and writes the result to rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x79.

Instruction Fields: Byte 0 holds opcode 0x79. Byte 1 bits 7-3 select destination integer register rd; byte 1 bits 2-0 and bytes 2-7 are reserved by this instruction and ignored by the processor. The value read is FPSR.

Exceptions: None.

Notes: None.

74. FMOVCR - fmovcr rd

Operation: rd = FPCR.

Assembler Syntax: fmovcr rd.

Attributes: Memory: none. Operand size: 32-bit control word. Privilege: unprivileged. Integer operand: rd receives FPCR, with writes to R0 discarded. FPSR/FPCR: reads FPCR; FPSR is not read or written.

Description: The processor evaluates FPCR and writes the result to rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x7A.

Instruction Fields: Byte 0 holds opcode 0x7A. Byte 1 bits 7-3 select destination integer register rd; byte 1 bits 2-0 and bytes 2-7 are reserved by this instruction and ignored by the processor. The value read is FPCR.

Exceptions: None.

Notes: None.

75. FMOVSC - fmovsc rs

Operation: FPSR = rs (masked).

Assembler Syntax: fmovsc rs.

Attributes: Memory: none. Operand size: 32-bit status word. Privilege: unprivileged. Integer operand: rs supplies the new FPSR value. FPSR/FPCR: writes FPSR after applying the FPSR writable-bit mask; FPCR is not read or written.

Description: The processor writes the architecturally writable status bits from rs into FPSR.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x7B.

Instruction Fields: Byte 0 holds opcode 0x7B. Byte 2 bits 7-3 select source integer register rs; bytes 1, byte 2 bits 2-0, and bytes 3-7 are reserved by this instruction and ignored by the processor. The destination is FPSR.

Exceptions: None.

Notes: None.

76. FMOVCC - fmovcc rs

Operation: FPCR = rs.

Assembler Syntax: fmovcc rs.

Attributes: Memory: none. Operand size: 32-bit control word. Privilege: unprivileged. Integer operand: rs supplies the new FPCR value. FPSR/FPCR: writes FPCR; FPSR is not read or written.

Description: The processor writes rs to FPCR subject to that control register's architectural write rules.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x7C.

Instruction Fields: Byte 0 holds opcode 0x7C. Byte 2 bits 7-3 select source integer register rs; bytes 1, byte 2 bits 2-0, and bytes 3-7 are reserved by this instruction and ignored by the processor. The destination is FPCR.

Exceptions: None.

Notes: None.

FPSR (Status Register): - Bits 27:24 - Condition Codes (N, Z, I, NaN). Overwritten per instruction. - Bits 3:0 - Exception Flags (UE, OE, DZ, IO). Sticky (IEEE-754). - **FMOVSC** masks the input value to preserve only these valid bits; bits 23:4 are reserved and always read as zero.

FPCR (Control Register): - Bits 1:0 - Rounding Mode: - 00: Nearest (default) - 01: Toward Zero (truncate) - 10: Toward - Inf (floor) - 11: Toward +Inf (ceil) - **FMOVCC** stores the full 32-bit source value in FPCR. FPU arithmetic interprets only bits 1:0 as the rounding mode; bits 31:2 are preserved and have no defined effect.

4.6.6 Double-Precision (Register Pairs)

77. DMOV - dmov fd, fs

Operation: fd = fs.

Assembler Syntax: dmov fd, fs.

Attributes: Memory: none. Operand size: double precision. Privilege: unprivileged. FP operands: fd and fs must encode even registers from f0 through f14. FPSR/FPCR: neither FPSR nor FPCR is read or written.

Description: The processor evaluates fs and writes the result to fd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x80.

Instruction Fields: Byte 0 holds opcode 0x80. Byte 1 bits 7-3 select destination FP register fd; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register fs; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid or odd fd or fs encoding enters the stopped processor state with PC unchanged.

Notes: None.

78. DLOAD - dload fd, disp(rs)

Operation: $fd = mem64[rs + disp]$.

Assembler Syntax: dload fd, disp(rs).

Attributes: Memory: 64-bit read. Operand size: double precision. Privilege: unprivileged. FP operands: fd must encode an even register from f0 through f14. Integer operand: rs supplies the base address. FPSR/FPCR: writes FPSR condition-code bits from the loaded value; FPCR is not read.

Description: The processor evaluates $mem64[rs + disp]$ and writes the result to fd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x81.

Instruction Fields: Byte 0 holds opcode 0x81. Byte 1 bits 7-3 select destination FP register fd; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select integer base register rs; byte 2 bits 2-0 and byte 3 are reserved by this instruction and ignored by the processor. Bytes 4-7 hold signed imm32 displacement in little-endian order for the 64-bit memory transfer.

Exceptions: An invalid or odd fd encoding enters the stopped processor state with PC unchanged. Read faults are those defined by the CPU memory and MMU rules.

Notes: None.

79. DSTORE - dstore fs, disp(rs)

Operation: $mem64[rs + disp] = fs$.

Assembler Syntax: dstore fs, disp(rs).

Attributes: Memory: 64-bit write. Operand size: double precision. Privilege: unprivileged. FP operands: fs must encode an even register from f0 through f14. Integer operand: rs supplies the base address. FPSR/FPCR: neither FPSR nor FPCR is read or written.

Description: The processor evaluates fs and writes the result to $mem64[rs + disp]$.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x82.

Instruction Fields: Byte 0 holds opcode 0x82. Byte 1 bits 7-3 select source FP register fs; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select integer base register rs; byte 2 bits 2-0 and byte 3 are reserved by this instruction and ignored by the processor. Bytes 4-7 hold signed imm32 displacement in little-endian order for the 64-bit memory transfer.

Exceptions: An invalid or odd fs encoding enters the stopped processor state with PC unchanged. Write faults are those defined by the CPU memory and MMU rules.

Notes: None.

80. DADD - dadd fd, fs, ft

Operation: $fd = fs + ft$.

Assembler Syntax: dadd fd, fs, ft.

Attributes: Memory: none. Operand size: double precision. Privilege: unprivileged. FP operands: fd, fs, and ft must encode even registers from f0 through f14. FPSR/FPCR: writes FPSR condition-code bits from the result and may set

FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates $fs + ft$ and writes the result to fd .

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x83.

Instruction Fields: Byte 0 holds opcode 0x83. Byte 1 bits 7-3 select destination FP register fd ; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register fs ; byte 3 bits 7-3 select source FP register ft . Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid or odd fd , fs , or ft encoding enters the stopped processor state with PC unchanged.

Notes: None.

81. DSUB - dsub fd, fs, ft

Operation: $fd = fs - ft$.

Assembler Syntax: dsub fd, fs, ft .

Attributes: Memory: none. Operand size: double precision. Privilege: unprivileged. FP operands: fd , fs , and ft must encode even registers from $f0$ through $f14$. FPSR/FPCR: writes FPSR condition-code bits from the result and may set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates $fs - ft$ and writes the result to fd .

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x84.

Instruction Fields: Byte 0 holds opcode 0x84. Byte 1 bits 7-3 select destination FP register fd ; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register fs ; byte 3 bits 7-3 select source FP register ft . Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid or odd fd , fs , or ft encoding enters the stopped processor state with PC unchanged.

Notes: None.

82. DMUL - dmul fd, fs, ft

Operation: $fd = fs * ft$.

Assembler Syntax: dmul fd, fs, ft .

Attributes: Memory: none. Operand size: double precision. Privilege: unprivileged. FP operands: fd , fs , and ft must encode even registers from $f0$ through $f14$. FPSR/FPCR: writes FPSR condition-code bits from the result and may set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates $fs * ft$ and writes the result to fd .

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x85.

Instruction Fields: Byte 0 holds opcode 0x85. Byte 1 bits 7-3 select destination FP register fd ; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register fs ; byte 3 bits 7-3 select source FP register ft . Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid or odd fd , fs , or ft encoding enters the stopped processor state with PC unchanged.

Notes: None.

83. DDIV - ddiv fd, fs, ft

Operation: $fd = fs / ft$.

Assembler Syntax: ddiv fd, fs, ft.

Attributes: Memory: none. Operand size: double precision. Privilege: unprivileged. FP operands: fd, fs, and ft must encode even registers from f0 through f14. FPSR/FPCR: writes FPSR condition-code bits from the result and may set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates fs / ft and writes the result to fd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x86.

Instruction Fields: Byte 0 holds opcode 0x86. Byte 1 bits 7-3 select destination FP register fd; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register fs; byte 3 bits 7-3 select source FP register ft. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid or odd fd, fs, or ft encoding enters the stopped processor state with PC unchanged.

Notes: None.

84. DMOD - dmod fd, fs, ft

Operation: $fd = fmod(fs, ft)$.

Assembler Syntax: dmod fd, fs, ft.

Attributes: Memory: none. Operand size: double precision. Privilege: unprivileged. FP operands: fd, fs, and ft must encode even registers from f0 through f14. FPSR/FPCR: writes FPSR condition-code bits from the result and may set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates $fmod(fs, ft)$ and writes the result to fd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x87.

Instruction Fields: Byte 0 holds opcode 0x87. Byte 1 bits 7-3 select destination FP register fd; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register fs; byte 3 bits 7-3 select source FP register ft. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid or odd fd, fs, or ft encoding enters the stopped processor state with PC unchanged.

Notes: None.

85. DABS - dabs fd, fs

Operation: $fd = \lfloor fs \rfloor$.

Assembler Syntax: dabs fd, fs.

Attributes: Memory: none. Operand size: double precision. Privilege: unprivileged. FP operands: fd and fs must encode even registers from f0 through f14. FPSR/FPCR: writes FPSR condition-code bits from the result and does not set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates $\lfloor fs \rfloor$ and writes the result to fd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x88.

Instruction Fields: Byte 0 holds opcode 0x88. Byte 1 bits 7-3 select destination FP register *fd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register *fs*; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid or odd *fd* or *fs* encoding enters the stopped processor state with PC unchanged.

Notes: None.

86. DNEG - *dneg fd, fs*

Operation: $fd = -fs$.

Assembler Syntax: *dneg fd, fs*.

Attributes: Memory: none. Operand size: double precision. Privilege: unprivileged. FP operands: *fd* and *fs* must encode even registers from *f0* through *f14*. FPSR/FPCR: writes FPSR condition-code bits from the result and does not set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates $-fs$ and writes the result to *fd*.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x89.

Instruction Fields: Byte 0 holds opcode 0x89. Byte 1 bits 7-3 select destination FP register *fd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register *fs*; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid or odd *fd* or *fs* encoding enters the stopped processor state with PC unchanged.

Notes: None.

87. DSQRT - *dsqrt fd, fs*

Operation: $fd = \sqrt{fs}$.

Assembler Syntax: *dsqrt fd, fs*.

Attributes: Memory: none. Operand size: double precision. Privilege: unprivileged. FP operands: *fd* and *fs* must encode even registers from *f0* through *f14*. FPSR/FPCR: writes FPSR condition-code bits from the result and may set FPSR sticky exception flags; FPCR is not read.

Description: The processor evaluates $\sqrt{fs}$ and writes the result to *fd*.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x8A.

Instruction Fields: Byte 0 holds opcode 0x8A. Byte 1 bits 7-3 select destination FP register *fd*; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register *fs*; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid or odd *fd* or *fs* encoding enters the stopped processor state with PC unchanged.

Notes: None.

88. DINT - dint fd, fs

Operation: $fd = \text{round}(fs)$.

Assembler Syntax: `dint fd, fs`.

Attributes: Memory: none. Operand size: double precision. Privilege: unprivileged. FP operands: `fd` and `fs` must encode even registers from `f0` through `f14`. FPSR/FPCR: reads FPCR rounding bits, writes FPSR condition-code bits from the rounded result, and does not set FPSR sticky exception flags.

Description: The processor evaluates $\text{round}(fs)$ and writes the result to `fd`.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x8B.

Instruction Fields: Byte 0 holds opcode 0x8B. Byte 1 bits 7-3 select destination FP register `fd`; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register `fs`; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid or odd `fd` or `fs` encoding enters the stopped processor state with PC unchanged.

Notes: None.

89. DCMF - dcmp rd, fs, ft

Operation: $rd = -1/0/1$.

Assembler Syntax: `dcmp rd, fs, ft`.

Attributes: Memory: none. Operand size: double precision. Privilege: unprivileged. FP operands: `fs` and `ft` must encode even registers from `f0` through `f14`; `rd` receives the integer comparison result. FPSR/FPCR: writes FPSR condition-code bits and may set the IO sticky flag for unordered operands; FPCR is not read.

Description: The processor compares the double-precision values in `fs:fs+1` and `ft:ft+1`, writes -1, 0, or 1 to `rd`, and updates FPSR condition state. If either operand is NaN, the comparison is unordered: `rd` receives 0, the NaN condition code is set, and the IO sticky exception flag is set.

Condition Codes: See the FPU condition-state description for this instruction class.

Instruction Format: Fixed 8-byte instruction; opcode = 0x8C.

Instruction Fields: Byte 0 holds opcode 0x8C. Byte 1 bits 7-3 select destination integer register `rd`; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register `fs`; byte 3 bits 7-3 select source FP register `ft`. Byte 2 bits 2-0, byte 3 bits 2-0, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid or odd `fs` or `ft` encoding enters the stopped processor state with PC unchanged.

Notes: +Inf compares equal to +Inf and greater than finite values.

90. DCVTIF - dcvtif fd, rs

Operation: $fd = \text{float64}(\text{int64}(rs))$.

Assembler Syntax: `dcvtif fd, rs`.

Attributes: Memory: none. Operand size: double precision result from a 64-bit signed integer source. Privilege: unprivileged. FP operands: `fd` must encode an even register from `f0` through `f14`; `rs` is an integer source register. FPSR/FPCR: writes FPSR condition-code bits from the FP result; FPCR is not read.

Description: The processor evaluates `float64(int64(rs))` and writes the result to `fd`.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x8D.

Instruction Fields: Byte 0 holds opcode 0x8D. Byte 1 bits 7-3 select destination FP register `fd`; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source integer register `rs`; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid or odd `fd` encoding enters the stopped processor state with PC unchanged.

Notes: None.

91. DCVTFI - `dcvtfi rd, fs`

Operation: `rd = int64(fs)` (saturating).

Assembler Syntax: `dcvtfi rd, fs`.

Attributes: Memory: none. Operand size: double precision source to 64-bit signed integer result. Privilege: unprivileged. FP operands: `fs` must encode an even register from `f0` through `f14`; `rd` receives the integer result. FPSR/FPCR: may set FPSR IO for NaN or saturation; FPCR is not read.

Description: The processor converts `fs` to a signed 64-bit integer with saturation and writes the result to `rd`.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x8E.

Instruction Fields: Byte 0 holds opcode 0x8E. Byte 1 bits 7-3 select destination integer register `rd`; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register `fs`; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid or odd `fs` encoding enters the stopped processor state with PC unchanged.

Notes: None.

92. FCVTSD - `fcvtzd fd, fs`

Operation: `fd = float64(float32(fs))`.

Assembler Syntax: `fcvtzd fd, fs`.

Attributes: Memory: none. Operand sizes: single precision source, double precision result. Privilege: unprivileged. FP operands: `fs` encodes `f0-f15`; `fd` must encode an even register from `f0` through `f14`. FPSR/FPCR: writes FPSR condition-code bits from the result; FPCR is not read.

Description: The processor evaluates `float64(float32(fs))` and writes the result to `fd`.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x8F.

Instruction Fields: Byte 0 holds opcode 0x8F. Byte 1 bits 7-3 select destination FP register `fd`; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register `fs`; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid `fs` encoding, or an invalid or odd `fd` encoding, enters the stopped processor state with PC unchanged.

Notes: None.

93. FCVTDS - fcvt ds fd, fs

Operation: $fd = \text{float32}(\text{float64}(fs))$.

Assembler Syntax: fcvt ds fd, fs.

Attributes: Memory: none. Operand sizes: double precision source, single precision result. Privilege: unprivileged. FP operands: fs must encode an even register from f0 through f14; fd encodes f0-f15. FPSR/FPCR: writes FPSR condition-code bits from the result; FPCR is not read.

Description: The processor evaluates $\text{float32}(\text{float64}(fs))$ and writes the result to fd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x90.

Instruction Fields: Byte 0 holds opcode 0x90. Byte 1 bits 7-3 select destination FP register fd; byte 1 bits 2-0 are reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select source FP register fs; byte 2 bits 2-0, byte 3, and bytes 4-7 are reserved by this instruction and ignored by the processor.

Exceptions: An invalid fd encoding, or an invalid or odd fs encoding, enters the stopped processor state with PC unchanged.

Notes: None.

Notes: - dload/dstore always transfer 8 bytes. - dcvt fi saturates to INT64_MAX/INT64_MIN on overflow and sets IO. - fcvt ds requires an even destination. fcvt ds requires an even source. - Double-precision opcodes are unsized; size suffixes are not used.

4.7 Branches

All branches are PC-relative. The branch offset is stored as a signed 32-bit value in the imm32 field. The new PC is calculated as:

$$PC_{\text{new}} = PC_{\text{current}} + \text{signExtend32to64}(\text{offset})$$

If the branch is not taken, PC advances by 8 (one instruction).

94. BRA - bra label

Operation: $PC = PC + \text{signExtend32to64}(\text{imm32})$.

Assembler Syntax: bra label.

Attributes: Condition: Always; Comparison: --.

Description: Unconditionally branches to the signed 32-bit PC-relative target encoded in imm32.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x40.

Instruction Fields: Byte 0 holds opcode 0x40. Bytes 1-3 are reserved by this instruction and ignored by the processor. Bytes 4-7 hold signed imm32 PC-relative branch offset in little-endian order.

Exceptions: None.

Notes: None.

95. BEQ - beq Rs, Rt, label

Operation: if $R_s == R_t$ then $PC = PC + \text{signExtend32to64}(\text{imm32})$ else $PC += 8$.

Assembler Syntax: beq Rs, Rt, label.

Attributes: Condition: $R_s == R_t$; Comparison: Unsigned (equality).

Description: Compares Rs and Rt for equality and branches to the signed 32-bit PC-relative target when the condition is true.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x41.

Instruction Fields: Byte 0 holds opcode 0x41. Byte 1 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select comparison register Rs; byte 3 bits 7-3 select comparison register Rt. Byte 2 bits 2-0 and byte 3 bits 2-0 are reserved. Bytes 4-7 hold signed imm32 PC-relative branch offset in little-endian order.

Exceptions: None.

Notes: None.

96. BNE - bne Rs, Rt, label

Operation: if $R_s != R_t$ then $PC = PC + \text{signExtend32to64}(\text{imm32})$ else $PC += 8$.

Assembler Syntax: bne Rs, Rt, label.

Attributes: Condition: $R_s != R_t$; Comparison: Unsigned (equality).

Description: Compares Rs and Rt for equality and branches to the signed 32-bit PC-relative target when the condition is true.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x42.

Instruction Fields: Byte 0 holds opcode 0x42. Byte 1 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select comparison register Rs; byte 3 bits 7-3 select comparison register Rt. Byte 2 bits 2-0 and byte 3 bits 2-0 are reserved. Bytes 4-7 hold signed imm32 PC-relative branch offset in little-endian order.

Exceptions: None.

Notes: None.

97. BLT - blt Rs, Rt, label

Operation: if $\text{int64}(R_s) < \text{int64}(R_t)$ then $PC = PC + \text{signExtend32to64}(\text{imm32})$ else $PC += 8$.

Assembler Syntax: blt Rs, Rt, label.

Attributes: Condition: $\text{int64}(R_s) < \text{int64}(R_t)$; Comparison: Signed.

Description: Interprets Rs and Rt as signed 64-bit integers and branches to the signed 32-bit PC-relative target when the condition is true.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x43.

Instruction Fields: Byte 0 holds opcode 0x43. Byte 1 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select comparison register Rs; byte 3 bits 7-3 select comparison register Rt. Byte 2 bits 2-0 and byte 3 bits 2-0 are reserved. Bytes 4-7 hold signed imm32 PC-relative branch offset in little-endian order.

Exceptions: None.

Notes: None.

98. BGE - bge Rs, Rt, label

Operation: if int64(Rs) >= int64(Rt) then PC = PC + signExtend32to64(imm32) else PC += 8.

Assembler Syntax: bge Rs, Rt, label.

Attributes: Condition: int64(Rs) >= int64(Rt); Comparison: Signed.

Description: Interprets Rs and Rt as signed 64-bit integers and branches to the signed 32-bit PC-relative target when the condition is true.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x44.

Instruction Fields: Byte 0 holds opcode 0x44. Byte 1 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select comparison register Rs; byte 3 bits 7-3 select comparison register Rt. Byte 2 bits 2-0 and byte 3 bits 2-0 are reserved. Bytes 4-7 hold signed imm32 PC-relative branch offset in little-endian order.

Exceptions: None.

Notes: None.

99. BGT - bgt Rs, Rt, label

Operation: if int64(Rs) > int64(Rt) then PC = PC + signExtend32to64(imm32) else PC += 8.

Assembler Syntax: bgt Rs, Rt, label.

Attributes: Condition: int64(Rs) > int64(Rt); Comparison: Signed.

Description: Interprets Rs and Rt as signed 64-bit integers and branches to the signed 32-bit PC-relative target when the condition is true.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x45.

Instruction Fields: Byte 0 holds opcode 0x45. Byte 1 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select comparison register Rs; byte 3 bits 7-3 select comparison register Rt. Byte 2 bits 2-0 and byte 3 bits 2-0 are reserved. Bytes 4-7 hold signed imm32 PC-relative branch offset in little-endian order.

Exceptions: None.

Notes: None.

100. BLE - ble Rs, Rt, label

Operation: if int64(Rs) <= int64(Rt) then PC = PC + signExtend32to64(imm32) else PC += 8.

Assembler Syntax: ble Rs, Rt, label.

Attributes: Condition: int64(Rs) <= int64(Rt); Comparison: Signed.

Description: Interprets Rs and Rt as signed 64-bit integers and branches to the signed 32-bit PC-relative target when the condition is true.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x46.

Instruction Fields: Byte 0 holds opcode 0x46. Byte 1 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select comparison register Rs; byte 3 bits 7-3 select comparison register Rt. Byte 2 bits 2-0 and byte 3 bits 2-0 are reserved. Bytes 4-7 hold signed imm32 PC-relative branch offset in little-endian order.

Exceptions: None.

Notes: None.

101. BHI - bhi Rs, Rt, label

Operation: if Rs > Rt then PC = PC + signExtend32to64(imm32) else PC += 8.

Assembler Syntax: bhi Rs, Rt, label.

Attributes: Condition: Rs > Rt; Comparison: Unsigned.

Description: Interprets Rs and Rt as unsigned 64-bit integers and branches to the signed 32-bit PC-relative target when the condition is true.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x47.

Instruction Fields: Byte 0 holds opcode 0x47. Byte 1 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select comparison register Rs; byte 3 bits 7-3 select comparison register Rt. Byte 2 bits 2-0 and byte 3 bits 2-0 are reserved. Bytes 4-7 hold signed imm32 PC-relative branch offset in little-endian order.

Exceptions: None.

Notes: None.

102. BLS - bls Rs, Rt, label

Operation: if Rs <= Rt then PC = PC + signExtend32to64(imm32) else PC += 8.

Assembler Syntax: bls Rs, Rt, label.

Attributes: Condition: Rs <= Rt; Comparison: Unsigned.

Description: Interprets Rs and Rt as unsigned 64-bit integers and branches to the signed 32-bit PC-relative target when the condition is true.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x48.

Instruction Fields: Byte 0 holds opcode 0x48. Byte 1 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select comparison register Rs; byte 3 bits 7-3 select comparison register Rt. Byte 2 bits 2-0 and byte 3 bits 2-0 are reserved. Bytes 4-7 hold signed imm32 PC-relative branch offset in little-endian order.

Exceptions: None.

Notes: None.

103. JMP - jmp (Rs) / jmp disp(Rs)

Operation: $PC = Rs + \text{signExtend32to64}(\text{imm32})$.

Assembler Syntax: jmp (Rs) / jmp disp(Rs).

Attributes: Condition: $PC = Rs + \text{signExtend}(\text{disp})$; Comparison: Register-indirect.

Description: Transfers control to the effective address computed from Rs plus the signed 32-bit displacement.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x49.

Instruction Fields: Byte 0 holds opcode 0x49. Byte 1 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select base register Rs; byte 2 bits 2-0 and byte 3 are reserved by this instruction and ignored by the processor. Bytes 4-7 hold signed imm32 displacement in little-endian order.

Exceptions: None.

Notes: None.

Encoding note for conditional branches: Rs is in byte 2[7:3], Rt is in byte 3[7:3], and the branch offset is in bytes 4-7 (imm32). The Rd field (byte 1[7:3]) is unused and should be encoded as 0. Branch targets are limited to the signed 32-bit PC-relative range described in section 7.3.

BRA uses only the imm32 field. Rs and Rt fields are unused.

JMP (opcode 0x49): - Computes the effective address as $\text{uint64}(\text{int64}(Rs) + \text{int64}(\text{int32}(\text{imm32})))$. - Transfers control to the effective address. - Does not modify the stack. No return address is saved. - Rs is in byte 2[7:3], the optional displacement is in bytes 4-7 (imm32). - Enables computed jumps, jump tables, and register-indirect branching.

4.8 Subroutine / Stack

104. JSR - jsr label

Operation: $SP -= 8$; $\text{mem}[SP] = PC + 8$; $PC = PC + \text{offset}$.

Assembler Syntax: jsr label.

Attributes: Memory: Write; Size: Q.

Description: The processor pushes $PC + 8$ on the stack and loads $PC + \text{offset}$ into PC.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x50.

Instruction Fields: Byte 0 holds opcode 0x50. Bytes 1-3 are reserved by this instruction and ignored by the processor. Bytes 4-7 hold signed imm32 PC-relative subroutine offset in little-endian order.

Exceptions: If MMU translation is enabled, the stack write can trap with cause 0 (FAULT_NOT_PRESENT), cause 2 (FAULT_WRITE_DENIED), or cause 10 (FAULT_SKAC). A physical 8-byte stack write outside implemented CPU-visible memory enters the stopped processor state and does not create a trap frame.

Notes: None.

105. RTS - rts

Operation: $PC = \text{mem}[SP]$; $SP += 8$.

Assembler Syntax: rts.

Attributes: Memory: Read; Size: Q.

Description: The processor pops an address from the stack into PC and increments SP by eight bytes.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x51.

Instruction Fields: Byte 0 holds opcode 0x51. Bytes 1-7 are reserved by this instruction and ignored by the processor.

Exceptions: If MMU translation is enabled, the stack read can trap with cause 0 (FAULT_NOT_PRESENT), cause 1 (FAULT_READ_DENIED), or cause 10 (FAULT_SKAC). A physical 8-byte stack read outside implemented CPU-visible memory enters the stopped processor state and does not create a trap frame.

Notes: None.

106. PUSH - push Rs

Operation: $SP -= 8$; $mem[SP] = Rs$.

Assembler Syntax: push Rs.

Attributes: Memory: Write; Size: Q.

Description: The processor decrements SP by eight bytes and stores Rs at the new stack top.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x52.

Instruction Fields: Byte 0 holds opcode 0x52. Byte 2 bits 7-3 select source register Rs; bytes 1, byte 2 bits 2-0, and bytes 3-7 are reserved by this instruction and ignored by the processor.

Exceptions: If MMU translation is enabled, the stack write can trap with cause 0 (FAULT_NOT_PRESENT), cause 2 (FAULT_WRITE_DENIED), or cause 10 (FAULT_SKAC). A physical 8-byte stack write outside implemented CPU-visible memory enters the stopped processor state and does not create a trap frame.

Notes: None.

107. POP - pop Rd

Operation: $Rd = mem[SP]$; $SP += 8$.

Assembler Syntax: pop Rd.

Attributes: Memory: Read; Size: Q.

Description: The processor loads a 64-bit value from the stack top into Rd and increments SP by eight bytes.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x53.

Instruction Fields: Byte 0 holds opcode 0x53. Byte 1 bits 7-3 select destination register Rd; byte 1 bits 2-0 and bytes 2-7 are reserved by this instruction and ignored by the processor.

Exceptions: If MMU translation is enabled, the stack read can trap with cause 0 (FAULT_NOT_PRESENT), cause 1 (FAULT_READ_DENIED), or cause 10 (FAULT_SKAC). A physical 8-byte stack read outside implemented CPU-visible memory enters the stopped processor state and does not create a trap frame.

Notes: None.

108. JSR - jsr (Rs) / jsr disp(Rs)

Operation: $SP -= 8$; $mem[SP] = PC + 8$; $PC = Rs + signExtend(disp)$.

Assembler Syntax: jsr (Rs) / jsr disp(Rs).

Attributes: Memory: Write; Size: Q.

Description: The processor pushes $PC + 8$ on the stack and loads $Rs + signExtend(disp)$ into PC.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0x54.

Instruction Fields: Byte 0 holds opcode 0x54. Byte 1 is reserved by this instruction and ignored by the processor. Byte 2 bits 7-3 select the base register Rs; byte 2 bits 2-0 and byte 3 are reserved by this instruction and ignored by the processor. Bytes 4-7 hold the signed displacement in little-endian order. Memory access class: Write.

Exceptions: If MMU translation is enabled, the stack write can trap with cause 0 (FAULT_NOT_PRESENT), cause 2 (FAULT_WRITE_DENIED), or cause 10 (FAULT_SKAC). A physical 8-byte stack write outside implemented CPU-visible memory enters the stopped processor state and does not create a trap frame.

Notes: None.

4.9 System

109. NOP - nop

Operation: No operation; $PC += 8$.

Assembler Syntax: nop.

Attributes: Memory: N; Size: --.

Description: No architectural register or memory state is modified; execution continues at the next instruction.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0xE0.

Instruction Fields: Byte 0 holds opcode 0xE0. Bytes 1-7 are reserved by this instruction and ignored by the processor.

Exceptions: None.

Notes: None.

110. HALT - halt

Operation: Stops execution.

Assembler Syntax: halt.

Attributes: Memory: N; Size: --.

Description: HALT enters the stopped processor state. The program counter is not advanced after the HALT word is fetched.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0xE1.

Instruction Fields: Byte 0 holds opcode 0xE1. Bytes 1-7 are reserved by this instruction and ignored by the processor.

Exceptions: None. No trap is generated.

Notes: HALT does not advance PC.

111. SEI - sei

Operation: Enable interrupts (set TIMER_CTRL bit 1).

Assembler Syntax: sei.

Attributes: Memory: N; Size: --.

Description: The processor enables interrupt delivery by setting the interrupt-enable bit in the timer control state.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0xE2.

Instruction Fields: Byte 0 holds opcode 0xE2. Bytes 1-7 are reserved by this instruction and ignored by the processor.

Exceptions: None.

Notes: None.

112. CLI - cli

Operation: Disable interrupts (clear TIMER_CTRL bit 1).

Assembler Syntax: cli.

Attributes: Memory: N; Size: --.

Description: The processor disables interrupt delivery by clearing the interrupt-enable bit in the timer control state.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0xE3.

Instruction Fields: Byte 0 holds opcode 0xE3. Bytes 1-7 are reserved by this instruction and ignored by the processor.

Exceptions: None.

Notes: None.

113. RTI - rti

Operation: Return from interrupt.

Assembler Syntax: rti.

Attributes: Memory: Read; Size: Q.

Description: The processor restores interrupt return state and resumes execution at the interrupted address.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0xE4.

Instruction Fields: Byte 0 holds opcode 0xE4. Bytes 1-7 are reserved by this instruction and ignored by the processor.

Exceptions: If MMU translation is enabled, the stack read can trap with cause 0 (FAULT_NOT_PRESENT), cause 1 (FAULT_READ_DENIED), or cause 10 (FAULT_SKAC). A physical 8-byte stack read outside implemented CPU-visible memory enters the stopped processor state and does not create a trap frame.

Notes: None.

114. WAIT - wait #usec

Operation: Sleep for imm32 microseconds; PC += 8.

Assembler Syntax: wait #usec.

Attributes: Memory: N; Size: --.

Description: The processor waits for the number of microseconds encoded in imm32 and then resumes at the next instruction.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0xE5.

Instruction Fields: Byte 0 holds opcode 0xE5. Bytes 1-3 are reserved by this instruction and ignored by the processor. Bytes 4-7 hold unsigned imm32 in little-endian order; this field is the requested wait interval in microseconds.

Exceptions: None.

Notes: None.

4.10 MMU, Privilege, and Atomic Instructions

115. MTCR - mtc CRn, Rs

Operation: CRn = Rs.

Assembler Syntax: mtc CRn, Rs.

Attributes: Privileged; Memory: N; Size: Q.

Description: The processor writes the 64-bit value in Rs to the selected control register. CRn is a 5-bit encoded field; assigned control registers are listed in section 8.1.1. Writing CR0 (PTBR) or changing CR5 bit 0 (MMU enable) invalidates all TLB entries. Writes to CR5 bit 1 (supervisor mode) and bit 4 (SUA latch) are ignored; those state bits are changed only by trap entry, ERET, SUAEN, and SUADIS. Supervisor-mode writes to unassigned control-register numbers have no architectural effect.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0xE6.

Instruction Fields: Byte 0 holds opcode 0xE6. Byte 1 bits 7-3 hold the control-register number CRn; byte 1 bits 2-0 are reserved. Byte 2 bits 7-3 select source register Rs; byte 2 bits 2-0 are reserved. Bytes 3-7 are reserved by this instruction and ignored by the processor.

Exceptions: User-mode execution raises FAULT_PRIV (cause 5). Writing CR15 (RAM_SIZE_BYTES) raises FAULT_ILLEGAL_INSTRUCTION (cause 11).

Notes: CR13 (PREV_MODE) is read-only by effect and ignores writes. CR3 (FAULT_PC) is writable so a trap handler can redirect the return address before ERET. Encodings CR16 through CR31 are reserved; MTCR to those encodings is ignored after the privilege check succeeds.

116. MFCR - mfcr Rd, CRn

Operation: Rd = CRn.

Assembler Syntax: mfcr Rd, CRn.

Attributes: Privileged except CR6; Memory: N; Size: Q.

Description: The processor reads the selected control register and writes the 64-bit value to Rd. CRn is a 5-bit encoded field; assigned control registers are listed in section 8.1.1. Register R0 remains hardwired to zero; an MFCR with Rd = R0 discards the value. Supervisor-mode reads of unassigned control-register numbers return zero.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0xE7.

Instruction Fields: Byte 0 holds opcode 0xE7. Byte 1 bits 7-3 select destination register Rd; byte 1 bits 2-0 are reserved. Byte 2 bits 7-3 hold the control-register number CRn; byte 2 bits 2-0 are reserved. Bytes 3-7 are reserved by this instruction and ignored by the processor.

Exceptions: User-mode execution raises FAULT_PRIV (cause 5) unless CRn is CR6 (TP). Reading CR6 is permitted in user mode.

Notes: User-mode exception: MFCR is normally supervisor-only, but reading CR6 (TP) is permitted in user mode. Reading CR5 composes the live MMU-enable, supervisor, SKEF, SKAC, and SUA bits. Reading CR15 returns the active CPU-visible RAM size. Encodings CR16 through CR31 are reserved; MFCR from those encodings returns zero after the privilege check succeeds.

117. ERET - eret

Operation: PC = CR3; restore the saved privilege and trap-frame state.

Assembler Syntax: eret.

Attributes: Privileged; Memory: N; Size: --.

Description: The processor consumes the active trap frame, sets PC to CR3 (FAULT_PC), and restores privilege state. If the interrupted context was user mode, ERET saves the current R31 to CR8 (KSP), loads R31 from CR12 (USP), enters user mode, enables interrupts, and clears the live SUA latch. If the interrupted context was supervisor mode, the processor remains supervisor and restores the live SUA latch from CR14 (SAVED_SUA).

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0xE8.

Instruction Fields: Byte 0 holds opcode 0xE8. Bytes 1-7 are reserved by this instruction and ignored by the processor.

Exceptions: User-mode execution raises FAULT_PRIV (cause 5).

Notes: ERET does not pop the data stack. The pop described here is the architectural trap-frame stack in section 11.14.

118. TLBFLUSH - tlbflush

Operation: Invalidate every TLB entry.

Assembler Syntax: tlbflush.

Attributes: Privileged; Memory: N; Size: --.

Description: The processor invalidates all entries in the 64-entry translation lookaside buffer. Subsequent translated fetches and data accesses perform page-table walks until new entries are cached.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0xE9.

Instruction Fields: Byte 0 holds opcode 0xE9. Bytes 1-7 are reserved by this instruction and ignored by the processor.

Exceptions: User-mode execution raises FAULT_PRIV (cause 5).

Notes: Execute TLBFLUSH after bulk page-table modifications. MTCR writes that change PTBR or MMU enable also invalidate all TLB entries.

119. TLBINVAL - tlbinal Rs

Operation: Invalidate the TLB entry selected by Rs >> 12.

Assembler Syntax: tlbinal Rs.

Attributes: Privileged; Memory: N; Size: Q.

Description: The processor treats Rs as a virtual address, derives the virtual page number by shifting it right by 12, and invalidates the matching TLB entry. Bits 11-0 of the address are ignored for invalidation.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0xEA.

Instruction Fields: Byte 0 holds opcode 0xEA. Byte 2 bits 7-3 select source register Rs; all other bits in bytes 1-7 are reserved by this instruction and ignored by the processor.

Exceptions: User-mode execution raises FAULT_PRIV (cause 5).

Notes: Rs contains an address within the affected virtual page, not a pre-shifted page number.

120. SYSCALL - syscall #imm32

Operation: CR1 = imm32; CR2 = 6; CR3 = PC + 8; PC = CR4.

Assembler Syntax: syscall #imm32.

Attributes: Any privilege level; Memory: N; Size: --.

Description: The processor enters the trap mechanism synchronously. It saves the syscall number from the immediate field in CR1 (FAULT_ADDR), saves cause code 6 in CR2 (FAULT_CAUSE), saves the address of the following instruction in CR3 (FAULT_PC), switches to supervisor mode, and transfers control to CR4 (TRAP_VEC).

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0xEB.

Instruction Fields: Byte 0 holds opcode 0xEB. Bytes 4-7 hold unsigned imm32 in little-endian order. Bytes 1-3 are reserved by this instruction and ignored by the processor.

Exceptions: None. SYSCALL is itself a trap source and records FAULT_SYSCALL (cause 6).

Notes: ERET from a syscall trap resumes at the instruction after SYSCALL unless the handler rewrites CR3.

121. SMODE - smode Rd

Operation: Rd = supervisor ? 1 : 0.

Assembler Syntax: smode Rd.

Attributes: Any privilege level; Memory: N; Size: Q.

Description: The processor writes 1 to Rd when the current privilege level is supervisor and 0 when the current privilege level is user. Register R0 remains hardwired to zero; SMODE R0 discards the result.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0xEC.

Instruction Fields: Byte 0 holds opcode 0xEC. Byte 1 bits 7-3 select destination register Rd; byte 1 bits 2-0 are reserved. Bytes 2-7 are reserved by this instruction and ignored by the processor.

Exceptions: None.

Notes: SMODE reads the current mode only; privilege changes occur on trap entry and ERET.

122. CAS - cas Rd, disp(Rs), Rt

Operation: old = mem64[addr]; if old == Rd then mem64[addr] = Rt; Rd = old.

Assembler Syntax: cas Rd, disp(Rs), Rt or cas Rd, (Rs), Rt.

Attributes: Atomic 64-bit read-modify-write; sequentially consistent; full memory barrier.

Description: The processor compares the 64-bit memory value at addr with the original value of Rd. If they match, Rt is written to memory. Rd receives the old memory value regardless of whether the swap occurs.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Atomic memory RMW form; opcode = 0xED.

Instruction Fields: Byte 0 holds opcode 0xED. Byte 1 bits 7-3 select Rd, which supplies the comparison value before the operation and receives the old memory value after it. Byte 2 bits 7-3 select base register Rs. Byte 3 bits 7-3 select operand register Rt, the replacement value. Bytes 4-7 hold signed imm32 displacement in little-endian order; addr = Rs + signExtend32to64(imm32).

Exceptions: FAULT_MISALIGNED for an unaligned effective address or for an address in a non-atomic CPU address aperture. MMU translation faults use their normal cause codes. After optional MMU translation, the physical 8-byte word must lie entirely inside the processor's atomic RAM aperture; a translated address outside that aperture raises FAULT_NOT_PRESENT.

Notes: Use the returned old value in Rd to test whether the compare-and-swap succeeded.

123. XCHG - xchg Rd, disp(Rs), Rt

Operation: old = mem64[addr]; mem64[addr] = Rt; Rd = old.

Assembler Syntax: xchg Rd, disp(Rs), Rt or xchg Rd, (Rs), Rt.

Attributes: Atomic 64-bit read-modify-write; sequentially consistent; full memory barrier.

Description: The processor atomically replaces the 64-bit memory value at addr with Rt and returns the old memory value in Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Atomic memory RMW form; opcode = 0xEE.

Instruction Fields: Byte 0 holds opcode 0xEE. Byte 1 bits 7-3 select Rd, which receives the old memory value. Byte 2 bits 7-3 select base register Rs. Byte 3 bits 7-3 select operand register Rt, the replacement value. Bytes 4-7 hold signed imm32 displacement in little-endian order; addr = Rs + signExtend32to64(imm32).

Exceptions: FAULT_MISALIGNED for an unaligned effective address or for an address in a non-atomic CPU address aperture. MMU translation faults use their normal cause codes. After optional MMU translation, the physical 8-byte word must lie entirely inside the processor's atomic RAM aperture; a translated address outside that aperture raises FAULT_NOT_PRESENT.

Notes: No size suffix is defined; the operation is always 64-bit.

124. FAA - faa Rd, disp(Rs), Rt

Operation: `old = mem64[addr]; mem64[addr] = old + Rt; Rd = old.`

Assembler Syntax: `faa Rd, disp(Rs), Rt` or `faa Rd, (Rs), Rt.`

Attributes: Atomic 64-bit read-modify-write; sequentially consistent; full memory barrier.

Description: The processor atomically adds `Rt` to the 64-bit memory value at `addr` and returns the old memory value in `Rd`.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Atomic memory RMW form; opcode = 0xEF.

Instruction Fields: Byte 0 holds opcode 0xEF. Byte 1 bits 7-3 select `Rd`, which receives the old memory value. Byte 2 bits 7-3 select base register `Rs`. Byte 3 bits 7-3 select operand register `Rt`, the addend. Bytes 4-7 hold signed `imm32` displacement in little-endian order; `addr = Rs + signExtend32to64(imm32)`.

Exceptions: `FAULT_MISALIGNED` for an unaligned effective address or for an address in a non-atomic CPU address aperture. MMU translation faults use their normal cause codes. After optional MMU translation, the physical 8-byte word must lie entirely inside the processor's atomic RAM aperture; a translated address outside that aperture raises `FAULT_NOT_PRESENT`.

Notes: Arithmetic wraps modulo 64 bits.

125. FAND - fand Rd, disp(Rs), Rt

Operation: `old = mem64[addr]; mem64[addr] = old & Rt; Rd = old.`

Assembler Syntax: `fand Rd, disp(Rs), Rt` or `fand Rd, (Rs), Rt.`

Attributes: Atomic 64-bit read-modify-write; sequentially consistent; full memory barrier.

Description: The processor atomically ANDs `Rt` with the 64-bit memory value at `addr` and returns the old memory value in `Rd`.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Atomic memory RMW form; opcode = 0xF0.

Instruction Fields: Byte 0 holds opcode 0xF0. Byte 1 bits 7-3 select `Rd`, which receives the old memory value. Byte 2 bits 7-3 select base register `Rs`. Byte 3 bits 7-3 select operand register `Rt`, the bit mask. Bytes 4-7 hold signed `imm32` displacement in little-endian order; `addr = Rs + signExtend32to64(imm32)`.

Exceptions: `FAULT_MISALIGNED` for an unaligned effective address or for an address in a non-atomic CPU address aperture. MMU translation faults use their normal cause codes. After optional MMU translation, the physical 8-byte word must lie entirely inside the processor's atomic RAM aperture; a translated address outside that aperture raises `FAULT_NOT_PRESENT`.

Notes: No size suffix is defined; the operation is always 64-bit.

126. FOR - for Rd, disp(Rs), Rt

Operation: `old = mem64[addr]; mem64[addr] = old | Rt; Rd = old.`

Assembler Syntax: `for Rd, disp(Rs), Rt` or `for Rd, (Rs), Rt.`

Attributes: Atomic 64-bit read-modify-write; sequentially consistent; full memory barrier.

Description: The processor atomically ORs Rt with the 64-bit memory value at addr and returns the old memory value in Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Atomic memory RMW form; opcode = 0xF1.

Instruction Fields: Byte 0 holds opcode 0xF1. Byte 1 bits 7-3 select Rd, which receives the old memory value. Byte 2 bits 7-3 select base register Rs. Byte 3 bits 7-3 select operand register Rt, the bit mask. Bytes 4-7 hold signed imm32 displacement in little-endian order; $\text{addr} = \text{Rs} + \text{signExtend32to64}(\text{imm32})$.

Exceptions: FAULT_MISALIGNED for an unaligned effective address or for an address in a non-atomic CPU address aperture. MMU translation faults use their normal cause codes. After optional MMU translation, the physical 8-byte word must lie entirely inside the processor's atomic RAM aperture; a translated address outside that aperture raises FAULT_NOT_PRESENT.

Notes: The mnemonic is FOR; it is unrelated to structured-language loop syntax.

127. FXOR - fxor Rd, disp(Rs), Rt

Operation: $\text{old} = \text{mem64}[\text{addr}]; \text{mem64}[\text{addr}] = \text{old} \wedge \text{Rt}; \text{Rd} = \text{old}.$

Assembler Syntax: fxor Rd, disp(Rs), Rt or fxor Rd, (Rs), Rt.

Attributes: Atomic 64-bit read-modify-write; sequentially consistent; full memory barrier.

Description: The processor atomically XORs Rt with the 64-bit memory value at addr and returns the old memory value in Rd.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Atomic memory RMW form; opcode = 0xF2.

Instruction Fields: Byte 0 holds opcode 0xF2. Byte 1 bits 7-3 select Rd, which receives the old memory value. Byte 2 bits 7-3 select base register Rs. Byte 3 bits 7-3 select operand register Rt, the bit mask. Bytes 4-7 hold signed imm32 displacement in little-endian order; $\text{addr} = \text{Rs} + \text{signExtend32to64}(\text{imm32})$.

Exceptions: FAULT_MISALIGNED for an unaligned effective address or for an address in a non-atomic CPU address aperture. MMU translation faults use their normal cause codes. After optional MMU translation, the physical 8-byte word must lie entirely inside the processor's atomic RAM aperture; a translated address outside that aperture raises FAULT_NOT_PRESENT.

Notes: No size suffix is defined; the operation is always 64-bit.

128. SUAEN - suaen

Operation: $\text{SUA} = 1.$

Assembler Syntax: suaen.

Attributes: Privileged; Memory: N; Size: --.

Description: The processor sets the supervisor-user-access latch, CR5 bit 4. When SKAC is enabled, this latch permits supervisor data accesses to pages marked user-accessible until the latch is cleared.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0xF3.

Instruction Fields: Byte 0 holds opcode 0xF3. Bytes 1-7 are reserved by this instruction and ignored by the processor.

Exceptions: User-mode execution raises FAULT_PRIV (cause 5).

Notes: Use SUADIS to close the same supervisor-user-access window.

129. SUADIS - suadis

Operation: SUA = 0.

Assembler Syntax: suadis.

Attributes: Privileged; Memory: N; Size: --.

Description: The processor clears the supervisor-user-access latch, CR5 bit 4. Clearing an already-clear latch leaves architectural state unchanged.

Condition Codes: No integer condition-code register is modified.

Instruction Format: Fixed 8-byte instruction; opcode = 0xF4.

Instruction Fields: Byte 0 holds opcode 0xF4. Bytes 1-7 are reserved by this instruction and ignored by the processor.

Exceptions: User-mode execution raises FAULT_PRIV (cause 5).

Notes: SUADIS does not change SKEF or SKAC; it only clears the live SUA latch.

5. Architectural Instruction Idioms

The following idioms use only real IE64 instructions and architecturally defined register behaviour.

5.1 Absolute Address Formation

Since R0 is hardwired to zero, `lea Rd, disp(r0)` loads the sign-extended 32-bit displacement as an address value.

```
lea r5, $A0000(r0)    ; r5 = $A0000
```

5.2 Full-Width Constant Formation

A full 64-bit constant is formed with `MOVE.L` for the low half followed by `MOVT` for the high half.

```
move.l r3, #$CAFEBABE
movt   r3, #$DEADBEEF
```

5.3 Zero-Comparison Branches

R0 can be used as the zero operand in all compare-and-branch instructions.

```
beq r5, r0, loop      ; branch if r5 == 0
bne r5, r0, loop      ; branch if r5 != 0
blt r5, r0, loop      ; branch if int64(r5) < 0
```

6. Addressing Modes

The IE64 supports the following addressing modes:

Mode	Syntax	Description	Used By
Immediate	#imm	32-bit immediate value, zero-extended to 64 bits	MOVE, ALU ops, FMOVECR, SYSCALL, WAIT
Register	Rs or Rt	Register contents (64-bit)	MOVE, ALU ops, branches, control-register moves
Register-indirect (data)	(Rs)	Memory at address in Rs	LOAD, STORE, FLOAD, FSTORE, DLOAD, DSTORE, atomics
Register-indirect (control)	(Rs)	Transfer control to address in Rs	JMP, JSR
Displacement	disp(Rs)	Address Rs + signExtend(disp)	LOAD, STORE, LEA, JMP, JSR, FLOAD, FSTORE, DLOAD, DSTORE, atomics
PC-relative	encoded signed imm32 offset	PC + signExtend(imm32)	BRA, Bcc, JSR

6.1 Immediate Addressing

The 32-bit immediate (imm32) is zero-extended to 64 bits when used as an operand (`operand3 = uint64(imm32)`). This means:

- Unsigned values 0 to 0xFFFFFFFF can be loaded directly.
- Negative 32-bit values can be formed with MOVEQ, which sign-extends imm32.
- The X bit (byte 1, bit 0) must be 1 to select immediate mode.

6.2 Displacement Addressing

Used by LOAD, STORE, LEA, JMP, JSR_IND, FLOAD, FSTORE, DLOAD, DSTORE, and atomic RMW instructions. The displacement is stored in imm32 and sign-extended to 64 bits before being added to the base register:

```
addr = uint64(int64(Rs) + int64(int32(imm32)))
```

This provides a +/- 2 GiB displacement range around the base register. The CPU keeps the effective virtual address at full 64-bit width through optional MMU translation and physical memory dispatch.

When imm32 is zero, the effective address is exactly the value in Rs. Control-transfer and memory instructions that encode a displacement consume the signed imm32 field directly; the CPU does not inspect the written source form.

7. Branch Architecture

7.1 Compare-and-Branch

The IE64 uses compare-and-branch instructions instead of a separate flags register. Each conditional branch instruction encodes:

- Two source registers (Rs, Rt) for comparison
- A signed 32-bit PC-relative offset

The comparison and branch are performed atomically in a single instruction. This eliminates the need for separate compare (CMP) and branch instructions, and avoids hazards associated with flag registers in pipelined CPUs.

7.2 Register-Indirect Transfer

JMP (opcode 0x49) provides register-indirect unconditional transfer. The target address is computed from a register plus an optional signed 32-bit displacement. This enables computed jumps, jump tables, and dispatch through register-held addresses.

JSR (opcode 0x54) provides the same register-indirect addressing for subroutine calls. It pushes the return address before transferring control, so a standard `rts` returns to the caller.

7.3 PC-Relative Offsets

All branch offsets are signed 32-bit values stored in the `imm32` field. The effective target address is:

```
target = PC + signExtend32to64(offset)
```

Where PC is the address of the branch instruction itself (not PC+8).

The instruction encoding stores a signed 32-bit offset. Targets outside the signed 32-bit PC-relative range are not representable by the branch encoding.

7.4 R0-as-Zero Idioms

Since R0 is hardwired to zero, comparisons against zero are natural:

Idiom	Instruction	Meaning
Branch if zero	<code>beq Rs, r0, label</code>	<code>Rs == 0</code>
Branch if nonzero	<code>bne Rs, r0, label</code>	<code>Rs != 0</code>
Branch if negative	<code>blt Rs, r0, label</code>	<code>int64(Rs) < 0</code>
Branch if non-negative	<code>bge Rs, r0, label</code>	<code>int64(Rs) >= 0</code>
Branch if positive	<code>bgt Rs, r0, label</code>	<code>int64(Rs) > 0</code>
Branch if non-positive	<code>ble Rs, r0, label</code>	<code>int64(Rs) <= 0</code>
Move zero to Rd	<code>move.q Rd, r0</code>	<code>Rd = 0</code>
Test equal to value	<code>move.q Rt, #val then beq Rs, Rt, label</code>	<code>Rs == val</code>

No condition-code register is required for these patterns.

7.5 Signed vs. Unsigned Comparisons

Branch	Comparison Type	Condition
BEQ	Equality (unsigned/signed irrelevant)	<code>Rs == Rt</code>
BNE	Equality (unsigned/signed irrelevant)	<code>Rs != Rt</code>
BLT	Signed	<code>int64(Rs) < int64(Rt)</code>
BGE	Signed	<code>int64(Rs) >= int64(Rt)</code>
BGT	Signed	<code>int64(Rs) > int64(Rt)</code>
BLE	Signed	<code>int64(Rs) <= int64(Rt)</code>
BHI	Unsigned	<code>uint64(Rs) > uint64(Rt)</code>

Branch	Comparison Type	Condition
BLS	Unsigned	uint64(Rs) <= uint64(Rt)

Note: For unsigned "greater than or equal" and "less than", use the complementary conditions: - Unsigned Rs >= Rt: Use bls Rt, Rs, label (swap operands and use BLS) or check with BHI + BEQ. - Unsigned Rs < Rt: Use bhi Rt, Rs, label (swap operands and use BHI).

8. Address Space and Reset State

8.1 CPU Address Space

IE64 defines 64-bit effective addresses. The programme counter, explicit LOAD/STORE effective addresses, stack addresses, FPU memory-operation addresses, atomic-operation addresses, and MMU translation inputs are all 64-bit quantities.

Property	Architectural definition
Effective-address width	64 bits
Data byte order	Little-endian
Instruction width	8 bytes
Integer memory widths	.B, .W, .L, .Q
Stack transfer width	8 bytes
Reset PC	PROG_START (0x1000)
Reset stack pointer	STACK_START (0x9F000)
Active RAM-size register	CR_RAM_SIZE_BYTES (CR15), read by MFCR

The IE64 CPU ISA defines effective-address behaviour and CPU-visible control state only. Addresses outside those CPU-visible rules have no additional meaning in this processor manual.

8.1.1 Control Register Numbers

Register	Number
CR_PTBR	CR0
CR_FAULT_ADDR	CR1
CR_FAULT_CAUSE	CR2
CR_FAULT_PC	CR3
CR_TRAP_VEC	CR4
CR_MMU_CTRL	CR5
CR_TP	CR6
CR_INTR_VEC	CR7
CR_KSP	CR8
CR_TIMER_PERIOD	CR9

Register	Number
CR_TIMER_COUNT	CR10
CR_TIMER_CTRL	CR11
CR_USP	CR12
CR_PREV_MODE	CR13
CR_SAVED_SUA	CR14
CR_RAM_SIZE_BYTES	CR15
Reserved	CR16-CR31

Unassigned control-register encodings are reserved. In supervisor mode, MFCR from a reserved encoding returns zero and MTCR to a reserved encoding has no effect. User-mode access still follows the privilege rule for the instruction: only MFCR CR6 is user-readable.

8.2 Initial State After Reset

Register/State	Value
PC	\$1000 (PROG_START)
R0	0 (hardwired)
R1-R29	0
R30	\$9F000
R31 (SP)	\$9F000 (STACK_START)
Interrupt enabled	false
In-interrupt flag	false
Timer enabled	false
Timer state	TIMER_STOPPED (0)
Timer count	0
Timer period	0

8.3 Stack

The stack grows downward from \$9F000 toward lower addresses. All stack operations (PUSH, POP, JSR, RTS, RTI, interrupt entry) use 8-byte (64-bit) transfers.

```
High memory:
$9F000 <-- Initial SP (STACK_START)
$9EFF8 <-- SP after first PUSH
$9EFF0 <-- SP after second PUSH
...
Low memory:
$01000 <-- PROG_START (stack must not grow past programme area)
```

9. Interrupt/Timer System

9.1 Non-MMU Fixed Interrupt Vector

In the non-MMU timer-interrupt path, IE64 uses a single fixed interrupt vector. This vector is initialised to 0 on reset. No instruction or externally addressable register changes the fixed vector. When the MMU is enabled and CR7 (INTR_VEC) is nonzero, timer interrupts use the CR7/ERET model described in section 9.4 and section 11.12 instead.

The vector table area at \$0000 is reserved.

9.2 Timer Registers

The architectural IE64 timer contract is the CR9/CR10/CR11 control-register interface:

Control register	Name	Architectural role
CR9	TIMER_PERIOD	Auto-reload value in timer-step units
CR10	TIMER_COUNT	Current countdown value in timer-step units
CR11	TIMER_CTRL	Enable and interrupt-enable bits

The IE64 timer is CPU-integrated. The architectural programming contract is the control-register interface above.

9.3 Timer Operation

The timer countdown advances once per decoded instruction step, after the instruction word and operand fields have been fetched and before the decoded instruction body is executed:

1. When the timer is enabled and TIMER_COUNT is nonzero, the CPU decrements the count by one for each decoded IE64 instruction step.
2. When the count reaches zero, timer state becomes expired.
3. If the timer is still enabled, the count is reloaded from TIMER_PERIOD.
4. If interrupts are enabled and the CPU is not already servicing an interrupt, the interrupt handler fires. A handler that reads TIMER_COUNT after entry observes the reloaded value when the timer remained enabled.
5. If TIMER_COUNT is zero and TIMER_PERIOD is nonzero at the next timer step, the count is loaded from TIMER_PERIOD and the timer returns to the running state.

9.4 Non-MMU Interrupt Flow

The following describes timer-interrupt entry when the MMU is disabled, or when INTR_VEC is zero:

1. The timer interrupt is delivered only when interrupt delivery is enabled and the interrupt-active latch is clear.
2. The interrupt-active latch is set, preventing nested delivery.
3. Push the current PC onto the stack: `SP -= 8; mem[SP] = PC.`
4. Set `PC = interruptVector.`
5. The ISR executes. It must end with RTI.
6. RTI pops the return address: `PC = mem[SP]; SP += 8.`
7. RTI clears the interrupt-active latch, re-enabling interrupt delivery.

When the MMU is enabled and INTR_VEC (CR7) is nonzero, the unified timer interrupt model is used (see section 11.12). Timer interrupts save PC to FAULT_PC, set FAULT_CAUSE=8, perform automatic stack switching only when the interrupted code was in user mode, and jump to INTR_VEC. The handler returns via ERET. When the MMU is off or

INTR_VEC is zero, the fixed-vector model is used: the CPU pushes PC and jumps to interruptVector, returning via RTI.

9.5 Interrupt Programming Patterns

9.5.1 Unified Model (MMU enabled)

When the MMU is enabled, timer interrupts can be delivered through INTR_VEC (CR7) using the ERET-based trap model. The handler uses the same return path as syscalls and faults:

```
; Kernel initialisation (supervisor mode)
move.l r1, #kernel_stack_top
mtcr cr8, r1          ; KSP = kernel stack
move.l r1, #timer_handler
mtcr cr7, r1          ; INTR_VEC = handler address
move.l r1, #44100
mtcr cr9, r1          ; TIMER_PERIOD = 44100 timer steps
move.l r1, #3
mtcr cr11, r1         ; TIMER_CTRL = enable + interrupt enable
; ... set up page table, enable MMU, ERET to user mode ...

timer_handler:
    push r1
    push r2
    ; ... handle timer interrupt ...
    pop r2
    pop r1
    eret              ; return to interrupted code (stack auto-restored)
```

9.5.2 Non-Programmable Fixed-Vector Model (MMU disabled)

The non-MMU push-PC/RTI delivery path uses the fixed reset vector and is not a programmable interrupt-vector ABI. On reset, interruptVector is zero, and there is no instruction that sets it. If a timer expires in this mode, delivery jumps to the fixed vector. Ordinary software cannot configure that vector.

Supervisor software that needs programmable IE64 timer interrupts should use the unified CR7/ERET model above: set KSP (CR8), INTR_VEC (CR7), TIMER_PERIOD (CR9), and TIMER_CTRL (CR11), then return from the handler with ERET.

9.6 SEI/CLI Semantics

Instruction	Effect
SEI	Sets TIMER_CTRL (CR11) bit 1, enabling interrupt delivery. The timer continues running; the next expiration after SEI will trigger an interrupt. Note: expirations that occurred while interrupts were disabled are not queued or replayed.
CLI	Clears TIMER_CTRL (CR11) bit 1, disabling interrupt delivery. Timer continues running and counting, but interrupts are not delivered.

Interrupts do not nest: while the interrupt-active latch is set, no new interrupt is delivered regardless of the interrupt-enable latch.

10. 64-bit Constant Loading

Because the imm32 field is only 32 bits wide, loading a full 64-bit constant requires two instructions.

10.1 Pattern: MOVE.L + MOV.T

```
; Load 0xDEADBEEF_CAFEBABE into R5
move.l r5, #$CAFEBABE      ; R5 = 0x00000000_CAFEBABE (lower 32 bits)
movt   r5, #$DEADBEEF      ; R5 = 0xDEADBEEF_CAFEBABE (upper 32 bits set)
```

Step by step: 1. `move.l r5, #$CAFEBABE` -- loads the 32-bit immediate `$CAFEBABE` into R5. Since this is a `.L` (32-bit) operation, the result is `0x00000000_CAFEBABE`. 2. `movt r5, #$DEADBEEF` -- takes the current value of R5, clears the upper 32 bits, and ORs in `$DEADBEEF << 32`. Result: `0xDEADBEEF_CAFEBABE`.

MOV.T operation: $Rd = (Rd \& 0x00000000FFFFFFFF) | (imm32 \ll 32)$

10.2 MOVEQ Alternative

For signed 32-bit values that need sign-extension to 64 bits:

```
moveq r5, #-1      ; R5 = 0xFFFFFFFF_FFFFFFFF (sign-extended from 32-bit -1)
moveq r5, #$80000000 ; R5 = 0xFFFFFFFF_80000000 (sign-extended)
```

MOVEQ interprets imm32 as a signed 32-bit integer and sign-extends it: $Rd = \text{int64}(\text{int32}(imm32))$.

11. Memory Management Unit

The IE64 includes an optional 6-level sparse radix paged MMU that provides virtual-to-physical address translation, page-level access control, and a supervisor/user privilege model. When enabled, instruction fetches and explicit memory operations are translated through a software-managed page table. The MMU is disabled on reset; supervisor code must build a page table and explicitly enable translation.

11.1 Privilege Levels

The IE64 operates in one of two privilege levels:

Level	MMU_CTRL.1	Description
Supervisor	1	Privileged execution mode. Can execute MTCR, MFCR, ERET, TLBFLUSH, TLBINVAL, SUAEN, and SUADIS. Page U-bit checks do not restrict supervisor access unless SKEF or SKAC is enabled.
User	0	Restricted. Privileged instructions cause a fault (cause code 5), except MFCR <code>Rd</code> , <code>CR6</code> which is user-readable. Can only access pages with U=1.

On reset the CPU is in supervisor mode. Transitioning to user mode is done via ERET (which clears supervisor mode and jumps to `FAULT_PC`). Returning to supervisor mode occurs only via a trap (fault or `SYSCALL`), which implicitly sets supervisor mode before jumping to the trap vector. The `SMODE` instruction reads the current mode into a register for introspection.

11.2 Control Registers

Sixteen control registers manage the MMU, thread state, timer, stack switching, trap state, and the live RAM-size discovery slot. They are accessed via MTCR (write) and MFCR (read).

CR	Name	R/W	Description
CR0	PTBR	RW	Page Table Base Register. Physical address of the page table.
CR1	FAULT_ADDR	RW	Virtual address that caused the most recent fault, or the syscall number (imm32) for SYSCALL. Writable so handlers can communicate information back. See 11.14 for the trap-stack semantics.
CR2	FAULT_CAUSE	RW	Cause code of the most recent fault (see 11.7). Writable for handler flexibility. See 11.14 for the trap-stack semantics.
CR3	FAULT_PC	RW	PC saved at trap entry. Used by ERET to resume. Writable: trap handlers must be able to modify this before ERET (e.g., to skip a faulting instruction or redirect execution). See 11.14 for the trap-stack semantics.
CR4	TRAP_VEC	RW	Trap handler entry PC. Jumped to on any fault or SYSCALL. When the MMU is enabled this PC is translated like any other instruction fetch.
CR5	MMU_CTRL	Special	Bit 0: MMU enable (RW). Bit 1: supervisor mode (RO). Bit 2: SKEF enable (RW). Bit 3: SKAC enable (RW). Bit 4: SUA latch (RO via MTCR; mutated only by SUAEN/SUADIS). See 11.2.1.
CR6	TP	RW	Thread Pointer. User-readable via MFCR (exception to the normal supervisor-only rule). Writable only in supervisor mode via MTCR. Intended for thread-local storage (TLS) base address.
CR7	INTR_VEC	RW	Timer interrupt handler PC. When MMU is enabled and INTR_VEC is nonzero, timer interrupts use the unified ERET-based entry model instead of the non-MMU push-PC/RTI model. The PC is translated like any other instruction fetch. Supervisor-only.
CR8	KSP	RW	Kernel Stack Pointer. Automatically swapped with R31 on user-to-supervisor transitions. Supervisor-only.
CR9	TIMER_PERIOD	RW	Timer reload period in decoded-instruction timer-step units. Supervisor-only.
CR10	TIMER_COUNT	RW	Current timer countdown value. Supervisor-only.
CR11	TIMER_CTRL	RW	Bit 0 = timer enable, Bit 1 = interrupt enable. SEI/CLI are aliases for setting/clearing bit 1. Supervisor-only.
CR12	USP	RW	Saved User Stack Pointer. Readable/writable in supervisor mode for context switch. Set automatically on user-to-supervisor transition. Supervisor-only.
CR13	PREV_MODE	RO	Previous privilege mode saved by <code>trapEntry</code> : 0 = trap came from user mode, 1 = trap came from supervisor mode. Read-only; set automatically on any trap/interrupt entry. Used by fault handlers to distinguish user faults (kill task) from kernel faults (panic). See 11.14 for the trap-stack semantics.
CR14	SAVED_SUA	RW	SUA latch snapshot taken on trap entry and consumed on ERET. Readable by kernel handlers that observe the interrupted code path's SUA state; writable so handlers can stage a custom value before ERET. See 11.2.1 and 11.14. Supervisor-only.
CR15	RAM_SIZE_BYTES	RO	Read-only active CPU-visible RAM size in bytes. Every MFCR observes the current architectural value. MTCR to CR15 raises <code>FAULT_ILLEGAL_INSTRUCTION</code> (cause 11). Supervisor-only.

PTBR must point to the physical address of the level-0 page table. The architecture does not require a special PTBR alignment beyond the byte addressability of physical memory, but supervisor software should allocate the level-0 table on a natural boundary and zero-initialise it before installing entries.

TRAP_VEC must be set before enabling the MMU, or faults will jump to address 0.

MMU_CTRL bit 0 is the master enable. Writing 1 activates translation for all subsequent memory accesses. Bit 1 (supervisor mode) is read-only; it reflects the current privilege level and cannot be written by MTCR. Bits 2-4 are the supervisor execute/access guard controls described below.

11.2.1 SKEF / SKAC / SUA (MMU_CTRL bits 2-4)

These bits are the IE64 supervisor execute/access guards.

Bit	Name	MTCR writable?	Description
2	SKEF	Yes	Supervisor-Kernel-Execute-Fault. When set, a supervisor instruction fetch from a page with PTE_U==1 raises FAULT_SKEF (cause 9).
3	SKAC	Yes	Supervisor-Kernel-Access-Check. When set, a supervisor read or write on a page with PTE_U==1 raises FAULT_SKAC (cause 10), unless the SUA latch is also set.
4	SUA	No (RO via MTCR)	Supervisor-User-Access latch. Mutated only by the privileged SUAEN and SUADIS opcodes. When set and SKAC is enabled, kernel data accesses to user pages succeed; when clear they fault with FAULT_SKAC .

Trap-entry / ERET discipline. On trap entry the SUA latch is snapshotted into CR_SAVED_SUA (CR14) and then forcibly cleared so a kernel handler cannot inherit an open supervisor-user-access window from the interrupted code path. On ERET the saved value is restored into the live latch when returning to supervisor mode (nested return); user-mode ERET clears the live latch unconditionally. See 11.14 for how the trap stack preserves CR_SAVED_SUA across nested traps automatically.

Helper idiom. Kernel user-memory access must bracket every supervisor load/store against a user pointer with SUAEN and SUADIS:

```

suaen                ; open the access window
load.b r3, (r1)      ; user load
store.b r3, (r2)     ; kernel store
suadis              ; close the access window

```

Kernel user-memory helpers must enforce this pattern around every supervisor access to user pages.

11.3 Page Table Format

The MMU uses a 6-level sparse radix page table for full 64-bit virtual addresses. Each page is 4 KiB, so virtual address bits 63:12 form a 52-bit VPN and bits 11:0 form the page offset.

Level layout:

Level	VPN Bits	Index Bits	Entries	Table Size
0 (top)	51..45	7	128	1 KiB
1	44..36	9	512	4 KiB
2	35..27	9	512	4 KiB
3	26..18	9	512	4 KiB
4	17..9	9	512	4 KiB
5 (leaf)	8..0	9	512	4 KiB

CR_PTBR points to the physical address of the level-0 table. Each entry is 8 bytes. Non-leaf entries use the same PTE encoding as leaf entries: the P bit marks the next-level table present, and the PPN field points to the next-level table.

Permission and A/D bits in non-leaf entries are ignored for non-leaf entries; the leaf entry controls access permissions.

11.4 Page Table Entry (PTE) Format

Bit: 63	12	11	7	6	5	4	3	2	1	0											
+	+	+	+	+	+	+	+	+	+	+											
	PPN (52 bits)						rsvd		D		A		U		X		W		R		P
+	+	+	+	+	+	+	+	+	+	+											

Bit(s)	Name	Description
0	P (Present)	Page is mapped. If P=0, any access faults with cause 0.
1	R (Read)	Read permission. If R=0, loads fault with cause 1.
2	W (Write)	Write permission. If W=0, stores fault with cause 2.
3	X (Execute)	Execute permission. If X=0, instruction fetch faults with cause 3.
4	U (User)	User-accessible. If U=0, user-mode access faults with cause 4.
5	A (Accessed)	Set by hardware on any successful translation (read, write, or execute).
6	D (Dirty)	Set by hardware on write access. Only set when the access is a store; reads and fetches do not set D.
11:7	--	Reserved; software should write 0. The MMU ignores these bits.
63:12	PPN	Physical Page Number (52 bits). Physical address = (PPN << 12) offset.

A/D bit semantics:

- The A and D bits are set by the MMU after all permission checks have passed. Both TLB-hit and TLB-miss paths perform A/D updates.
- A is set on every successful translation regardless of access type (read, write, execute).
- D is set only on write accesses.
- The bits are written back to the page table entry in memory only when they change (i.e., when the bit was previously 0). This avoids unnecessary memory writes on repeated accesses to the same page.
- Architectural constraint:** Page tables must reside in normal writable physical memory. The A/D write-back performs a physical store to the PTE; if the page table were placed over non-RAM storage, the write-back would not be a valid page-table update.
- Kernel software clears A/D bits by rewriting the PTE directly in memory and then flushing the TLB (via TLBFLUSH or TLBINVAL) to ensure the cached TLB entry is also updated. This is the basis for page reclamation and working-set estimation algorithms.

11.5 Virtual Address Format

Bit: 63	12	11	0
+-----+	+-----+		
	VPN (52 bits)		Offset (12 bits)
+-----+	+-----+		

Page count is no longer a fixed MMU_NUM_PAGES = 8192; software should derive relevant bounds from active visible RAM queried via CR_RAM_SIZE_BYTES.

11.6 MMU Instructions

Nine privilege and MMU opcodes are defined. All except SYSCALL and SMODE are privileged (supervisor-only); executing them in user mode faults with cause code 5 (privilege violation). MFCR has a special exception: reading CR6 (TP) is permitted in user mode.

Their instruction-by-instruction entries are part of the Complete Instruction Reference in section 4.9. This section defines the control-register and MMU state those instructions manipulate.

11.7 Trap Model

Traps are raised by faults (translation errors, permission violations) and by the SYSCALL instruction. On trap entry:

1. CR1 (FAULT_ADDR) is set to the faulting virtual address (for faults) or the syscall number (for SYSCALL).
2. CR2 (FAULT_CAUSE) is set to the cause code.
3. CR3 (FAULT_PC) is set to the relevant PC (see below).
4. If the trap came from user mode, automatic stack switching occurs: user R31 is saved to USP (CR12), R31 is loaded from KSP (CR8). Supervisor-origin traps do not swap stacks. See section 11.11.
5. The CPU switches to supervisor mode, or remains in supervisor mode for supervisor-origin traps.
6. PC is set to CR4 (TRAP_VEC).

Differentiated PC save: - **SYSCALL:** CR3 = PC + 8 (address of the instruction *after* SYSCALL). ERET resumes execution past the syscall. - **Faults:** CR3 = faulting PC (address of the instruction that caused the fault). ERET re-executes the faulting instruction after the handler fixes the page table.

This distinction means trap handlers do not need to adjust the return address; ERET always restores CR3 directly.

11.8 Fault Cause Codes

Code	Name	Trigger
0	Page Not Present	Absent PTE mapping or unavailable physical/atomic backing. Access to a page with P=0, access whose physical memory backing is unavailable, and an atomic operation outside the atomic RAM aperture all raise cause 0.
1	Read Denied	Load from a page with R=0.
2	Write Denied	Store to a page with W=0.
3	Execute Denied	Instruction fetch from a page with X=0.
4	User/Supervisor	User-mode access to a page with U=0.
5	Privilege Violation	User-mode execution of a privileged instruction (MTCR, MFCR except MFCR Rd, CR6, ERET, TLBFLUSH, TLBINVAL, SUAEN, SUADIS).
6	Syscall	SYSCALL instruction executed.
7	Misaligned	Atomic memory operation (CAS, XCHG, FAA, FAND, FOR, FXOR) with address not 8-byte aligned.
8	Timer Interrupt	Timer interrupt (delivered via INTR_VEC when MMU enabled).
9	SKEF	Supervisor instruction fetch from a page with PTE_U==1 while MMU_CTRL.SKEF is set.
10	SKAC	Supervisor data read or write on a page with PTE_U==1 while MMU_CTRL.SKAC is set and MMU_CTRL.SUA is clear.
11	Illegal Instruction	Opcode-level invariants the CPU cannot otherwise enforce. Currently raised only by MTCR RAM_SIZE_BYTES, Rs (the CR is read-only - see 11.2).

11.9 Translation Lookaside Buffer (TLB)

The MMU maintains a 64-entry direct-mapped translation lookaside buffer to cache page table lookups.

- **Indexing:** TLB[VPN & 63]. Each entry stores the VPN tag, physical page number, leaf-PTE physical address, and translation permission flags.
- **Lookup:** On every translated access, the TLB is checked first. On a hit (VPN tag matches), the cached translation and permission flags are used. On a miss, the page table is walked, the PTE is loaded, permission checks are performed, and the translation result is cached in the TLB.
- **Invalidation:** TLBFLUSH clears all 64 entries. TLBINVAL clears only the entry matching the given VA's VPN. Writing PTBR or MMU_CTRL via MTCR also flushes the entire TLB.
- **Coherency:** The TLB is not automatically coherent with page table memory. After modifying a PTE in memory, software must execute TLBINVAL for the affected page (or TLBFLUSH for bulk changes) before the new mapping takes effect.

11.10 W^X Page Policy

The IE64 page-table format supports a Write XOR Execute page policy by giving each page independent W and X permission bits:

- **Code pages:** P=1, X=1, W=0, with R optional. Execute-only user text (P=1, R=0, X=1) is a first-class contract.
- **Data/stack pages:** P=1, R=1, W=1, X=0 (readable + writable, not executable).

Supervisor page-table code should enforce the policy by avoiding PTEs that set both W and X. The MMU checks the requested access against the relevant bit but does not reject a PTE merely because both W and X are set. To load new code under a W^X policy, supervisor software maps the target pages as writable, writes the code, then remaps them as executable with appropriate TLB invalidation between steps.

11.11 Automatic Stack Switching

The IE64 provides automatic kernel/user stack separation via KSP (CR8) and USP (CR12). On any user-to-supervisor transition (SYSCALL, fault, or timer interrupt):

1. The previous privilege mode is saved in CPU trap state.
2. The current user R31 (stack pointer) is saved to USP (CR12).
3. R31 is loaded from KSP (CR8), giving the handler a kernel stack.
4. The CPU switches to supervisor mode.

On ERET:

1. If the previous mode was user: R31 is saved to KSP (CR8), R31 is restored from USP (CR12), and the CPU returns to user mode.
2. If the previous mode was supervisor: the CPU stays in supervisor mode with no stack swap.

This allows trap handlers to execute on a dedicated kernel stack without any software stack-switching prologue. The kernel must initialise KSP (via MTCR) before entering user mode for the first time.

11.12 Unified Timer Interrupt Model

The IE64 supports two timer interrupt models, selected automatically based on the MMU state and INTR_VEC (CR7):

Unified model (MMU enabled, INTR_VEC nonzero):

Timer interrupts are delivered through the same trap mechanism as faults and SYSCALLs:

1. PC is saved to CR3 (FAULT_PC).

2. CR2 (FAULT_CAUSE) is set to 8 (FAULT_TIMER).
3. If the interrupt interrupted user mode, automatic stack switching occurs (user R31 saved to USP, R31 loaded from KSP). Supervisor-origin timer interrupts do not swap stacks.
4. The CPU switches to supervisor mode, or remains in supervisor mode for supervisor-origin interrupts.
5. PC is set to INTR_VEC (CR7).

The handler returns via ERET, which restores the stack and privilege mode automatically. This model eliminates the need for RTI and provides a consistent trap-return path for all supervisor entry points.

Fixed-vector model (MMU disabled, or INTR_VEC is zero):

Timer interrupts use the classic push-PC/RTI mechanism:

1. The current PC is pushed onto the stack: `SP -= 8; mem[SP] = PC`.
2. The interrupt-active latch is set, preventing nested delivery.
3. PC is set to `interruptVector`.
4. The handler returns via RTI, which pops PC and clears the interrupt-active latch.

This model is the defined timer-interrupt path when MMU vectoring is not active.

11.13 Atomic Memory Operations

The IE64 provides six atomic read-modify-write (RMW) instructions for lock-free synchronisation. These instructions use a dedicated encoding form that repurposes all three register fields and the immediate field:

Encoding: Memory RMW Form

Byte:	0	1	2	3	4	5	6	7
	+-----+-----+-----+-----+-----+-----+-----+-----+							
	Opcode	Rd<<3 0 0	Rs<<3	Rt<<3		imm32 (LE)		
	+-----+-----+-----+-----+-----+-----+-----+-----+							

- **Rd**: Destination register (receives the old value read from memory)
- **Rs**: Base register (memory address source)
- **Rt**: Operand register (value to swap, add, or use in bitwise operation)
- **imm32**: Signed displacement added to Rs to form the effective address

Effective address: `addr = uint64(int64(Rs) + int64(int32(imm32)))`

Instruction-by-instruction entries for CAS, XCHG, FAA, FAND, FOR, and FXOR are part of the Complete Instruction Reference in section 4.9.

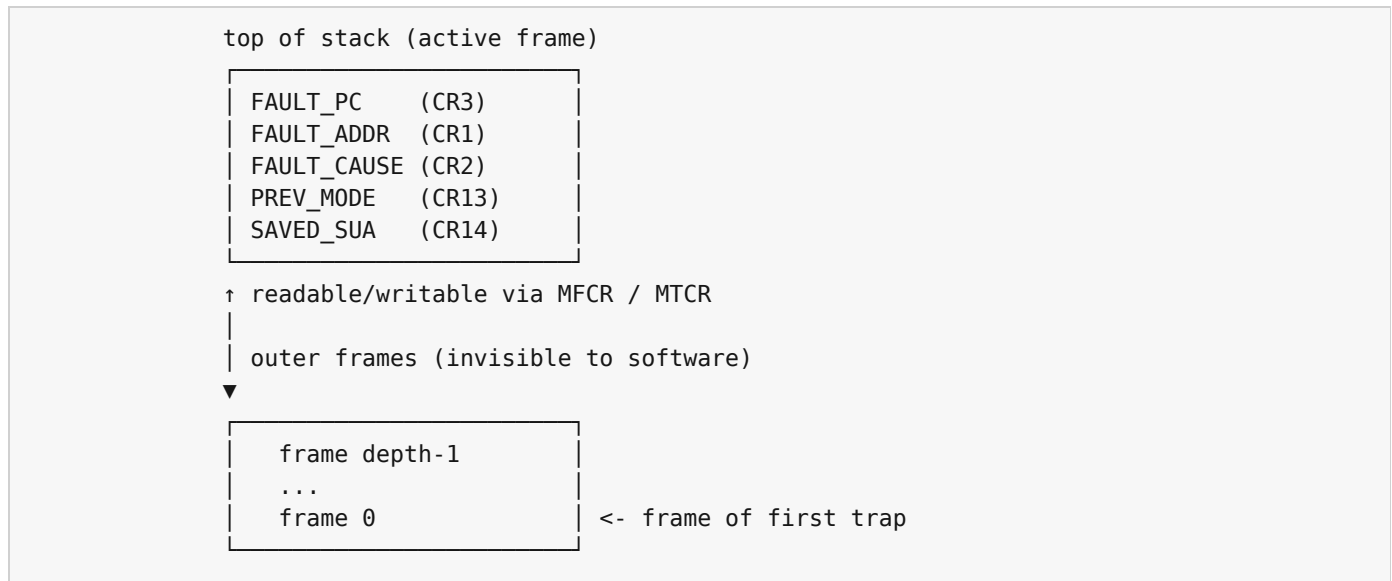
11.13.1 Shared Atomic Semantics

- **Size**: Always 64-bit (.Q). No size suffix is accepted; atomic operations operate on naturally-aligned 64-bit words only.
- **Alignment**: The effective address must be 8-byte aligned (`addr & 7 == 0`). A misaligned address causes a trap with FAULT_MISALIGNED (cause code 7).
- **Atomic RAM aperture only**: Atomic operations are valid only for aligned 64-bit words in the processor's atomic RAM aperture. An unaligned effective address or an address in a non-atomic CPU address aperture raises FAULT_MISALIGNED (cause 7). After optional MMU translation, the physical 8-byte word must lie entirely inside the processor's atomic RAM aperture. A translated address outside that aperture raises FAULT_NOT_PRESENT (cause 0).
- **Ordering**: All atomic operations are sequentially consistent. They act as full memory barriers.

- **MMU:** When the MMU is enabled, the effective address is translated as a write operation through the normal page-table translation path. A/D bits are set accordingly. The resulting physical word must lie inside the atomic RAM aperture; non-contiguous physical memory beyond that aperture is not a target for these instructions.
- **CAS (Compare-And-Swap):** Reads the 64-bit value at [addr] into a temporary. If the temporary equals the current value of Rd, the value of Rt is written to [addr]. Regardless of whether the swap occurred, Rd receives the old value from memory. This allows the caller to detect success by comparing the returned old value against the expected value.

11.14 Trap-Frame Stack

Nested-trap state preservation is architectural rather than kernel-managed. The CPU owns a fixed-depth trap-frame stack that holds the outer trap's FAULT_PC, PREV_MODE, CR_SAVED_SUA, FAULT_ADDR, and FAULT_CAUSE. The live CR fields remain the canonical "top of stack" accessed through MFCR / MTCR; the stack is not directly visible to software.



Push. On trap entry (fault, SYSCALL, timer interrupt) the CPU snapshots the active frame (all five fields) onto the stack before overwriting them. The snapshot happens first; subsequent trap-entry bookkeeping (setting PREV_MODE, saving and clearing the SUA latch into SAVED_SUA) then writes the new active-frame values.

Pop. On ERET the CPU consumes the active frame (uses FAULT_PC as the new PC, restores the SUA latch from SAVED_SUA or clears it on user return, and swaps stacks on user return) and then pops the previous frame off the stack into the active fields. When the stack is empty the active fields are cleared to zero, matching the fresh-boot state.

Overflow. The stack depth is fixed. Exceeding it halts the CPU with a trap-stack overflow condition: runaway nested faults are always a kernel bug and must be visible, not silently dropped.

Implications for kernel handlers. Handlers do **not** need to save and restore CR_FAULT_PC or CR_SAVED_SUA on the kernel stack to survive a nested synchronous trap. The trap stack preserves the outer context automatically. Existing kernel code that performs a manual MFCR CR_FAULT_PC / MTCR CR_FAULT_PC prologue around a possibly-faulting region still works - such save/restore now writes the active frame, so the restore writes back the same value already preserved. The older handler pattern is thus redundant but harmless; new handlers should omit it.

Reset. Processor reset clears the trap stack to depth 0 and clears all trap-frame slots. After reset, no saved trap frame is architecturally visible.

Appendix A: Opcode Map

A.1 Instruction Set Summary

Opcode	Operation	Syntax
MOVE	Data Movement	MOVE Rd, Rs / Rd, #imm
MOVT	Data Movement	MOVT Rd, #imm
MOVEQ	Data Movement	MOVEQ Rd, #imm
LEA	Data Movement	LEA Rd, disp(Rs)
LOAD	Memory Access	LOAD Rd, disp(Rs)
STORE	Memory Access	STORE Rd, disp(Rs)
ADD	Arithmetic	ADD Rd, Rs, Rt/#imm
SUB	Arithmetic	SUB Rd, Rs, Rt/#imm
MULU	Arithmetic	MULU Rd, Rs, Rt/#imm
MULS	Arithmetic	MULS Rd, Rs, Rt/#imm
DIVU	Arithmetic	DIVU Rd, Rs, Rt/#imm
DIVS	Arithmetic	DIVS Rd, Rs, Rt/#imm
MOD	Arithmetic	MOD Rd, Rs, Rt/#imm
NEG	Arithmetic	NEG Rd, Rs
MODS	Arithmetic	MODS Rd, Rs, Rt/#imm
MULHU	Arithmetic	MULHU Rd, Rs, Rt/#imm
MULHS	Arithmetic	MULHS Rd, Rs, Rt/#imm
AND	Logical	AND Rd, Rs, Rt/#imm
OR	Logical	OR Rd, Rs, Rt/#imm
EOR	Logical	EOR Rd, Rs, Rt/#imm
NOT	Logical	NOT Rd, Rs
LSL	Shift	LSL Rd, Rs, Rt/#imm
LSR	Shift	LSR Rd, Rs, Rt/#imm
ASR	Shift	ASR Rd, Rs, Rt/#imm
CLZ	Shift	CLZ.l Rd, Rs
SEXT	Shift	SEXT Rd, Rs
ROL	Shift	ROL Rd, Rs, Rt/#imm
ROR	Shift	ROR Rd, Rs, Rt/#imm
CTZ	Shift	CTZ.l Rd, Rs
POPCNT	Shift	POPCNT.l Rd, Rs
BSWAP	Shift	BSWAP.l Rd, Rs

Opcode	Operation	Syntax
BRA	Branch	BRA label
BEQ	Branch	BEQ Rs, Rt, label
BNE	Branch	BNE Rs, Rt, label
BLT	Branch	BLT Rs, Rt, label
BGE	Branch	BGE Rs, Rt, label
BGT	Branch	BGT Rs, Rt, label
BLE	Branch	BLE Rs, Rt, label
BHI	Branch	BHI Rs, Rt, label
BLS	Branch	BLS Rs, Rt, label
JMP	Branch	JMP (Rs) / disp(Rs)
JSR	Subroutine	JSR label
RTS	Subroutine	RTS
PUSH	Stack	PUSH Rs
POP	Stack	POP Rd
JSR	Subroutine	JSR (Rs) / disp(Rs)
FMOV	FPU	FMOV fd, fs
FLOAD	FPU	FLOAD fd, disp(rs)
FSTORE	FPU	FSTORE fs, disp(rs)
FADD	FPU	FADD fd, fs, ft
FSUB	FPU	FSUB fd, fs, ft
FMUL	FPU	FMUL fd, fs, ft
FDIV	FPU	FDIV fd, fs, ft
FMOD	FPU	FMOD fd, fs, ft
FABS	FPU	FABS fd, fs
FNEG	FPU	FNEG fd, fs
FSQRT	FPU	FSQRT fd, fs
FINT	FPU	FINT fd, fs
FCMP	FPU	FCMP rd, fs, ft
FCVTIF	FPU	FCVTIF fd, rs
FCVTFI	FPU	FCVTFI rd, fs
FMOVI	FPU	FMOVI fd, rs
FMOVO	FPU	FMOVO rd, fs
FSIN	FPU	FSIN fd, fs
FCOS	FPU	FCOS fd, fs

Opcode	Operation	Syntax
FTAN	FPU	FTAN fd, fs
FATAN	FPU	FATAN fd, fs
FLOG	FPU	FLOG fd, fs
FEXP	FPU	FEXP fd, fs
FPOW	FPU	FPOW fd, fs, ft
FMOVECR	FPU	FMOVECR fd, #idx
FMOVSR	FPU	FMOVSR rd
FMOVCR	FPU	FMOVCR rd
FMOVSC	FPU	FMOVSC rs
FMOVCC	FPU	FMOVCC rs
DMOV	FPU64	DMOV fd, fs
DLOAD	FPU64	DLOAD fd, disp(rs)
DSTORE	FPU64	DSTORE fs, disp(rs)
DADD	FPU64	DADD fd, fs, ft
DSUB	FPU64	DSUB fd, fs, ft
DMUL	FPU64	DMUL fd, fs, ft
DDIV	FPU64	DDIV fd, fs, ft
DMOD	FPU64	DMOD fd, fs, ft
DABS	FPU64	DABS fd, fs
DNEG	FPU64	DNEG fd, fs
DSQRT	FPU64	DSQRT fd, fs
DINT	FPU64	DINT fd, fs
DCMP	FPU64	DCMP rd, fs, ft
DCVTIF	FPU64	DCVTIF fd, rs
DCVTFI	FPU64	DCVTFI rd, fs
FCVTSD	FPU64	FCVTSD fd, fs
FCVTDS	FPU64	FCVTDS fd, fs
NOP	System	NOP
HALT	System	HALT
SEI	System	SEI
CLI	System	CLI
RTI	System	RTI
WAIT	System	WAIT #usec
MTCR	MMU	MTCR CRn, Rs

Opcode	Operation	Syntax
MFCR	MMU	MFCR Rd, CRn
ERET	MMU	ERET
TLBFLUSH	MMU	TLBFLUSH
TLBINVAL	MMU	TLBINVAL Rs
SYSCALL	MMU	SYSCALL #imm32
SMODE	MMU	SMODE Rd
CAS	Atomic	CAS Rd, disp(Rs), Rt
XCHG	Atomic	XCHG Rd, disp(Rs), Rt
FAA	Atomic	FAA Rd, disp(Rs), Rt
FAND	Atomic	FAND Rd, disp(Rs), Rt
FOR	Atomic	FOR Rd, disp(Rs), Rt
FXOR	Atomic	FXOR Rd, disp(Rs), Rt
SUAEN	System	SUAEN
SUADIS	System	SUADIS

A.2 Machine Opcode Encoding Map

Opcode Byte	Mnemonic	Category	Operands
0x01	MOVE	Data Movement	Rd, Rs / Rd, #imm
0x02	MOVT	Data Movement	Rd, #imm
0x03	MOVEQ	Data Movement	Rd, #imm
0x04	LEA	Data Movement	Rd, disp(Rs)
0x10	LOAD	Memory Access	Rd, disp(Rs)
0x11	STORE	Memory Access	Rd, disp(Rs)
0x20	ADD	Arithmetic	Rd, Rs, Rt/#imm
0x21	SUB	Arithmetic	Rd, Rs, Rt/#imm
0x22	MULU	Arithmetic	Rd, Rs, Rt/#imm
0x23	MULS	Arithmetic	Rd, Rs, Rt/#imm
0x24	DIVU	Arithmetic	Rd, Rs, Rt/#imm
0x25	DIVS	Arithmetic	Rd, Rs, Rt/#imm
0x26	MOD	Arithmetic	Rd, Rs, Rt/#imm
0x27	NEG	Arithmetic	Rd, Rs
0x28	MODS	Arithmetic	Rd, Rs, Rt/#imm
0x29	MULHU	Arithmetic	Rd, Rs, Rt/#imm
0x2A	MULHS	Arithmetic	Rd, Rs, Rt/#imm

Opcode Byte	Mnemonic	Category	Operands
0x30	AND	Logical	Rd, Rs, Rt/#imm
0x31	OR	Logical	Rd, Rs, Rt/#imm
0x32	EOR	Logical	Rd, Rs, Rt/#imm
0x33	NOT	Logical	Rd, Rs
0x34	LSL	Shift	Rd, Rs, Rt/#imm
0x35	LSR	Shift	Rd, Rs, Rt/#imm
0x36	ASR	Shift	Rd, Rs, Rt/#imm
0x37	CLZ	Shift	Rd, Rs
0x38	SEXT	Shift	Rd, Rs
0x39	ROL	Shift	Rd, Rs, Rt/#imm
0x3A	ROR	Shift	Rd, Rs, Rt/#imm
0x3B	CTZ	Shift	Rd, Rs
0x3C	POPCNT	Shift	Rd, Rs
0x3D	BSWAP	Shift	Rd, Rs
0x40	BRA	Branch	label
0x41	BEQ	Branch	Rs, Rt, label
0x42	BNE	Branch	Rs, Rt, label
0x43	BLT	Branch	Rs, Rt, label
0x44	BGE	Branch	Rs, Rt, label
0x45	BGT	Branch	Rs, Rt, label
0x46	BLE	Branch	Rs, Rt, label
0x47	BHI	Branch	Rs, Rt, label
0x48	BLS	Branch	Rs, Rt, label
0x49	JMP	Branch	(Rs) / disp(Rs)
0x50	JSR	Subroutine	label
0x51	RTS	Subroutine	(none)
0x52	PUSH	Stack	Rs
0x53	POP	Stack	Rd
0x54	JSR	Subroutine	(Rs) / disp(Rs)
0x60	FMOV	FPU	fd, fs
0x61	FLOAD	FPU	fd, disp(rs)
0x62	FSTORE	FPU	fs, disp(rs)
0x63	FADD	FPU	fd, fs, ft
0x64	FSUB	FPU	fd, fs, ft

Opcode Byte	Mnemonic	Category	Operands
0x65	FMUL	FPU	fd, fs, ft
0x66	FDIV	FPU	fd, fs, ft
0x67	FMOD	FPU	fd, fs, ft
0x68	FABS	FPU	fd, fs
0x69	FNEG	FPU	fd, fs
0x6A	FSQRT	FPU	fd, fs
0x6B	FINT	FPU	fd, fs
0x6C	FCMP	FPU	rd, fs, ft
0x6D	FCVTIF	FPU	fd, rs
0x6E	FCVTFI	FPU	rd, fs
0x6F	FMOVI	FPU	fd, rs
0x70	FMOVO	FPU	rd, fs
0x71	FSIN	FPU	fd, fs
0x72	FCOS	FPU	fd, fs
0x73	FTAN	FPU	fd, fs
0x74	FATAN	FPU	fd, fs
0x75	FLOG	FPU	fd, fs
0x76	FEXP	FPU	fd, fs
0x77	FPOW	FPU	fd, fs, ft
0x78	FMOVECR	FPU	fd, #idx
0x79	FMOVSR	FPU	rd
0x7A	FMOVCR	FPU	rd
0x7B	FMOVSC	FPU	rs
0x7C	FMOVCC	FPU	rs
0x80	DMOV	FPU64	fd, fs
0x81	DLOAD	FPU64	fd, disp(rs)
0x82	DSTORE	FPU64	fs, disp(rs)
0x83	DADD	FPU64	fd, fs, ft
0x84	DSUB	FPU64	fd, fs, ft
0x85	DMUL	FPU64	fd, fs, ft
0x86	DDIV	FPU64	fd, fs, ft
0x87	DMOD	FPU64	fd, fs, ft
0x88	DABS	FPU64	fd, fs
0x89	DNEG	FPU64	fd, fs

Opcode Byte	Mnemonic	Category	Operands
0x8A	DSQRT	FPU64	fd, fs
0x8B	DINT	FPU64	fd, fs
0x8C	DCMP	FPU64	rd, fs, ft
0x8D	DCVTIF	FPU64	fd, rs
0x8E	DCVTFI	FPU64	rd, fs
0x8F	FCVTSD	FPU64	fd, fs
0x90	FCVTDS	FPU64	fd, fs
0xE0	NOP	System	(none)
0xE1	HALT	System	(none)
0xE2	SEI	System	(none)
0xE3	CLI	System	(none)
0xE4	RTI	System	(none)
0xE5	WAIT	System	#usec
0xE6	MTCR	MMU	CRn, Rs
0xE7	MFCR	MMU	Rd, CRn
0xE8	ERET	MMU	(none)
0xE9	TLBFLUSH	MMU	(none)
0xEA	TLBINVAL	MMU	Rs
0xEB	SYSCALL	MMU	#imm32
0xEC	SMODE	MMU	Rd
0xED	CAS	Atomic	Rd, disp(Rs), Rt
0xEE	XCHG	Atomic	Rd, disp(Rs), Rt
0xEF	FAA	Atomic	Rd, disp(Rs), Rt
0xF0	FAND	Atomic	Rd, disp(Rs), Rt
0xF1	FOR	Atomic	Rd, disp(Rs), Rt
0xF2	FXOR	Atomic	Rd, disp(Rs), Rt
0xF3	SUAEN	System	(none)
0xF4	SUADIS	System	(none)

A.3 Opcode Ranges

Range	Category
\$01-\$04	Data Movement
\$10-\$11	Memory Access
\$20-\$2A	Arithmetic

Range	Category
\$30-\$3D	Logical / Shift
\$40-\$49	Branches
\$50-\$54	Subroutine / Stack
\$60-\$7C	Floating Point (FPU)
\$80-\$90	Double-precision Floating Point (FPU64)
\$E0-\$E5	System
\$E6-\$EC	MMU
\$ED-\$F2	Atomic Memory Operations
\$F3-\$F4	Supervisor-user-access controls

Opcodes not listed above are reserved. Executing a reserved opcode enters the stopped processor state without architecturally advancing PC.

Appendix B: Encoding Examples

B.1 `move.l r5, #$CAFEBAE`

```

Opcode = 0x01 (MOVE)
Rd = 5, Size = 2 (.L), X = 1 (immediate)
Rs = 0, Rt = 0
imm32 = 0xCAFEBAE

Byte 0: 0x01
Byte 1: (5 << 3) | (2 << 1) | 1 = 0x28 | 0x04 | 0x01 = 0x2D
Byte 2: 0 << 3 = 0x00
Byte 3: 0 << 3 = 0x00
Bytes 4-7: 0xBE 0xBA 0xFE 0xCA (little-endian)

Binary: 01 2D 00 00 BE BA FE CA

```

B.2 `add.q r3, r1, r2`

```

Opcode = 0x20 (ADD)
Rd = 3, Size = 3 (.Q), X = 0 (register)
Rs = 1, Rt = 2
imm32 = 0

Byte 0: 0x20
Byte 1: (3 << 3) | (3 << 1) | 0 = 0x18 | 0x06 | 0x00 = 0x1E
Byte 2: 1 << 3 = 0x08
Byte 3: 2 << 3 = 0x10
Bytes 4-7: 0x00 0x00 0x00 0x00

Binary: 20 1E 08 10 00 00 00 00

```

B.3 beq r1, r2, target (target 24 bytes ahead)

```
Opcode = 0x41 (BEQ)
Rd = 0 (unused), Size = 3 (.Q), X = 0
Rs = 1, Rt = 2
imm32 = 24 (signed offset = +24)

Byte 0: 0x41
Byte 1: (0 << 3) | (3 << 1) | 0 = 0x06
Byte 2: 1 << 3 = 0x08
Byte 3: 2 << 3 = 0x10
Bytes 4-7: 0x18 0x00 0x00 0x00

Binary: 41 06 08 10 18 00 00 00
```

B.4 store.b r7, 4(r10)

```
Opcode = 0x11 (STORE)
Rd = 7, Size = 0 (.B), X = 1 (displacement non-zero)
Rs = 10, Rt = 0
imm32 = 4

Byte 0: 0x11
Byte 1: (7 << 3) | (0 << 1) | 1 = 0x38 | 0x00 | 0x01 = 0x39
Byte 2: 10 << 3 = 0x50
Byte 3: 0x00
Bytes 4-7: 0x04 0x00 0x00 0x00

Binary: 11 39 50 00 04 00 00 00
```

B.5 push r15

```
Opcode = 0x52 (PUSH)
Rd = 0 (unused), Size = 3 (.Q), X = 0
Rs = 15, Rt = 0
imm32 = 0

Byte 0: 0x52
Byte 1: (0 << 3) | (3 << 1) | 0 = 0x06
Byte 2: 15 << 3 = 0x78
Byte 3: 0x00
Bytes 4-7: 0x00 0x00 0x00 0x00

Binary: 52 06 78 00 00 00 00 00
```

B.6 jmp (r5)

```
Opcode = 0x49 (JMP)
Rd = 0 (unused), Size = 0, X = 0
Rs = 5, Rt = 0
imm32 = 0 (no displacement)

Byte 0: 0x49
Byte 1: 0x00
Byte 2: 5 << 3 = 0x28
Byte 3: 0x00
Bytes 4-7: 0x00 0x00 0x00 0x00

Binary: 49 00 28 00 00 00 00 00
```

B.7 jsr 16(r3)

```
Opcode = 0x54 (JSR indirect)
Rd = 0 (unused), Size = 0, X = 0
Rs = 3, Rt = 0
imm32 = 16 (displacement)

Byte 0: 0x54
Byte 1: 0x00
Byte 2: 3 << 3 = 0x18
Byte 3: 0x00
Bytes 4-7: 0x10 0x00 0x00 0x00

Binary: 54 00 18 00 10 00 00 00
```